Cost-Weighted Correlated Gaussian Bandits: From Minimax Lower Bounds to Non-Asymptotic δ-PAC Soundness and First-Order Asymptotic Cost Optimality

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Abstract

We study fixed-confidence best-arm identification in a two-armed Gaussian bandit in which, besides pulling each arm individually, the learner may pull both arms jointly at cost $c_{1,2}$ and observe a bivariate Gaussian pair with unknown correlation ρ. We derive an instance-dependent lower bound on the expected sampling cost and solve the underlying max–min problem analytically. For symmetric individual costs the optimal cost allocation is always a vertex, either pure joint sampling or balanced individual sampling, with a phase transition at an explicit threshold $c^{\mathrm{th}}_{1,2}(\delta_0, \rho)$ determined by a scalar cubic equation; in particular, under strong positive correlation joint pulls remain optimal even when they cost more than two individual pulls. We derive the profiled generalized likelihood ratio (GLR) statistic in closed form, construct a data-dependent stopping threshold from a mixture-martingale argument and an explicit curvature bound that makes the stopping rule δ-PAC non-asymptotically, and prove that a cost-weighted correlated Track-and-Stop algorithm is first-order asymptotically cost-optimal. Experiments show cost reductions over Track-and-Stop, Top-Two Thompson Sampling, KL-LUCB and Successive Elimination when the correlation is strong, while matching the best baseline when joint pulls do not help.

Keywords: best-arm identification, fixed confidence, correlated bandits, cost-aware sampling, Track-and-Stop, generalized likelihood ratio test

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1 Introduction

In fixed-confidence best-arm identification, a learner samples arms sequentially and must stop and declare the arm with the largest mean, with error probability at most δ, while using as few resources as possible. Garivier and Kaufmann [2] characterized the optimal expected sample complexity through a max–min problem and showed that the Track-and-Stop algorithm attains it asymptotically as $\delta \to 0$. In many applications, however, samples carry different costs, and two alternatives can often be measured together on a shared sample. Joint measurements are then affected by common noise, which cancels in the difference between the two arms when the measurements are positively correlated.

This report studies the simplest such model: two Gaussian arms that can be pulled individually at cost c each, or jointly at cost $c_{1,2}$, where a joint pull returns a bivariate Gaussian pair with unknown correlation ρ. Our contributions are as follows.

- **Lower bound (Section 2).** We derive the instance-dependent lower bound on the expected stopping cost, $\mathrm{kl}(\delta, 1-\delta)/I^\star(\theta)$, where $I^\star(\theta)$ is defined in (5).
- **Analytical resolution of lower bound and phase transition (Section 3).** We solve the max–min problem in closed form, both for known and unknown correlation. The optimal allocation is a vertex of the cost simplex, and the switch between pure joint sampling and balanced individual sampling happens at an explicit threshold $c^{\mathrm{th}}_{1,2}(\delta_0, \rho)$. We also formulate the K-armed generalization as a semi-definite program.
- **Stopping statistic and δ-PAC threshold (Sections 4 and 5).** We derive the profiled GLR statistic in closed form and prove that a threshold built from a mixture martingale, an explicit curvature bound and a heavy-tailed prior gives a stopping rule that is δ-PAC for every δ, without any boundedness assumption on the means.
- **Asymptotic optimality (Section 6).** We prove that a cost-weighted correlated Track-and-Stop algorithm with forced joint exploration matches the lower bound as $\delta \to 0$.
- **Experiments (Section 7).** We confirm the predicted phase transition and compare against standard best-arm identification algorithms.

2 Problem Formulation and Lower Bound Calculation

2.1 The Cost-Weighted Correlated Bandit Environment

Let $\theta = (\mu_1, \mu_2, \rho) \in \Theta := \mathbb{R}^2 \times (-1, 1)$ denote the unknown environment parameter, where $\mu_1, \mu_2 \in \mathbb{R}$ are the arm means ($\mu_1 \neq \mu_2$) and $\rho \in (-1, 1)$ is the correlation between the arms when sampled jointly. When analyzing the lower bound and asymptotic optimality, we assume without loss of generality that Arm 1 is the unique best arm ($\mu_1 > \mu_2$), and we denote the true mean gap by $\Delta := \mu_1 - \mu_2 > 0$ (written simply as δ in Section 3 where no ambiguity with the confidence level arises).

At each discrete decision round $t \geq 1$, the learner selects one of three sampling actions $A_t \in \{1, 2, 12\}$:

- **Action 1 (Individual pull of Arm 1):** Incurs cost $c_1 > 0$ and yields a scalar observation $X_{1,i} \sim \mathcal{N}(\mu_1, 1)$.
- **Action 2 (Individual pull of Arm 2):** Incurs cost $c_2 > 0$ and yields a scalar observation $X_{2,i} \sim \mathcal{N}(\mu_2, 1)$.
- **Action 12 (Joint pull of Arms 1 and 2):** Incurs cost $c_{1,2} > 0$ and yields a bivariate Gaussian observation pair

$$\begin{pmatrix} U_j \\ V_j \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \Sigma \right), \qquad \text{with} \qquad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

Let $N_1(t)$, $N_2(t)$, and $N_{1,2}(t)$ denote the action counts after $t = N_1(t) + N_2(t) + N_{1,2}(t)$ total pulls. The cumulative measurement cost incurred up to round t is

$$\mathrm{Cost}(t) := c_1 N_1(t) + c_2 N_2(t) + c_{1,2} N_{1,2}(t), \qquad c_{\min} t \le \mathrm{Cost}(t) \le c_{\max} t, \tag{1}$$

where $c_{\min} := \min\{c_1, c_2, c_{1,2}\}$ and $c_{\max} := \max\{c_1, c_2, c_{1,2}\}$.

A fixed-confidence algorithm consists of a filtration-adapted sampling rule $(A_t)_{t\ge1}$, an almost-surely finite stopping time τ with respect to the observation history $\mathcal{F}_t := \sigma(A_1, Y_1, \ldots, A_t, Y_t)$, and a terminal recommendation rule $\widehat{a}_\tau \in \{1, 2\}$. An algorithm is called δ-PAC (Probably Approximately Correct at confidence level $\delta \in (0, 1)$) if for every $\theta = (\mu_1, \mu_2, \rho) \in \Theta$ with $\mu_1 \neq \mu_2$,

$$\mathbb{P}_\theta\Big(\tau < \infty, \widehat{a}_\tau \neq \arg\max_{i\in\{1,2\}} \mu_i\Big) \le \delta.$$

2.2 Explicit Kullback–Leibler Divergences

Let $\lambda = (\mu_1', \mu_2', \rho') \in \Theta$ be any candidate alternative parameter. The Kullback–Leibler (KL) divergences per pull of Actions 1, 2, and 12 between the observation distributions under θ and λ are computed as follows:

1. **Individual Pulls ($a \in \{1, 2\}$):** Between the univariate Gaussian distributions $\mathcal{N}(\mu_i, 1)$ and $\mathcal{N}(\mu_i', 1)$, the KL divergence is

$$D_1(\theta, \lambda) = \frac{(\mu_1 - \mu_1')^2}{2}, \qquad D_2(\theta, \lambda) = \frac{(\mu_2 - \mu_2')^2}{2}. \tag{2}$$

2. **Joint Pull ($a = 12$):** The true covariance matrix $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ has determinant $|\Sigma| = 1 - \rho^2$. The alternative covariance matrix $\Sigma' = \begin{pmatrix} 1 & \rho' \\ \rho' & 1 \end{pmatrix}$ has determinant $|\Sigma'| = 1 - \rho'^2$ and inverse

$$(\Sigma')^{-1} = \frac{1}{1-\rho'^2} \begin{pmatrix} 1 & -\rho' \\ -\rho' & 1 \end{pmatrix}.$$

Using the general formula for the KL divergence between two 2-dimensional Gaussian distributions,

$$D_{12}(\theta, \lambda) = \frac{1}{2}\left[\log\frac{|\Sigma'|}{|\Sigma|} - 2 + \mathrm{tr}\big((\Sigma')^{-1}\Sigma\big) + \begin{pmatrix}\mu_1' - \mu_1 & \mu_2' - \mu_2\end{pmatrix}(\Sigma')^{-1}\begin{pmatrix}\mu_1' - \mu_1 \\ \mu_2' - \mu_2\end{pmatrix}\right].$$

Evaluating the matrix product inside the trace gives

$$(\Sigma')^{-1}\Sigma = \frac{1}{1-\rho'^2}\begin{pmatrix} 1 & -\rho' \\ -\rho' & 1 \end{pmatrix}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = \frac{1}{1-\rho'^2}\begin{pmatrix} 1-\rho\rho' & \rho-\rho' \\ \rho-\rho' & 1-\rho\rho' \end{pmatrix},$$

so $\mathrm{tr}\big((\Sigma')^{-1}\Sigma\big) = \frac{2(1-\rho\rho')}{1-\rho'^2}$, and therefore

$$-2 + \mathrm{tr}\big((\Sigma')^{-1}\Sigma\big) = \frac{-2(1-\rho'^2) + 2(1-\rho\rho')}{1-\rho'^2} = \frac{2(\rho'^2 - \rho\rho')}{1-\rho'^2}.$$

Expanding the quadratic form in the means,

$$\begin{aligned}&\begin{pmatrix}\mu_1' - \mu_1 & \mu_2' - \mu_2\end{pmatrix}\frac{1}{1-\rho'^2}\begin{pmatrix} 1 & -\rho' \\ -\rho' & 1 \end{pmatrix}\begin{pmatrix}\mu_1' - \mu_1 \\ \mu_2' - \mu_2\end{pmatrix} \\ &\qquad = \frac{(\mu_1 - \mu_1')^2 - 2\rho'(\mu_1 - \mu_1')(\mu_2 - \mu_2') + (\mu_2 - \mu_2')^2}{1-\rho'^2}.\end{aligned}$$

Combining these terms yields the exact joint KL divergence:

$$D_{12}(\theta, \lambda) = \frac{1}{2}\log\frac{1-\rho'^2}{1-\rho^2} + \frac{\rho'^2 - \rho\rho'}{1-\rho'^2} + \frac{(\mu_1 - \mu_1')^2 - 2\rho'(\mu_1 - \mu_1')(\mu_2 - \mu_2') + (\mu_2 - \mu_2')^2}{2(1-\rho'^2)}. \tag{3}$$

2.3 Instance-Dependent Cost Lower Bound

For $i \in \{1,2\}$, let $\mathrm{Alt}(i) := \{(\mu_1', \mu_2', \rho') \in \Theta : \mu_i' \le \mu_{3-i}'\}$ denote the set of alternative parameters under which Arm i is not the unique optimal arm, and write $\mathrm{Alt}_1 := \mathrm{Alt}(1) = \{(\mu_1', \mu_2', \rho') \in \Theta : \mu_1' \le \mu_2'\}$.

Theorem 2.1 (Cost Lower Bound). *Fix $\theta = (\mu_1, \mu_2, \rho) \in \Theta$ with $\mu_1 > \mu_2$. For any δ-PAC algorithm $(\tau, \widehat{a}_\tau)$ satisfying $\mathbb{E}_\theta[\mathrm{Cost}(\tau)] < \infty$,*

$$\mathbb{E}_\theta[\mathrm{Cost}(\tau)] \ge \frac{\mathrm{kl}(\delta, 1-\delta)}{I^\star(\theta)}, \qquad \textit{and hence} \qquad \liminf_{\delta \downarrow 0} \frac{\mathbb{E}_\theta[\mathrm{Cost}(\tau)]}{\log(1/\delta)} \ge \frac{1}{I^\star(\theta)}, \tag{4}$$

where $\mathrm{kl}(x,y) := x \log \frac{x}{y} + (1-x) \log \frac{1-x}{1-y}$, *and*

$$I^\star(\theta) := \sup_{\substack{\gamma_1, \gamma_2, \gamma_{1,2} \ge 0 \\ c_1\gamma_1 + c_2\gamma_2 + c_{1,2}\gamma_{1,2} = 1}} \inf_{\lambda \in \mathrm{Alt}_1} \Big[\gamma_1 D_1(\theta, \lambda) + \gamma_2 D_2(\theta, \lambda) + \gamma_{1,2} D_{12}(\theta, \lambda) \Big]. \tag{5}$$

Here, $\gamma_a = \alpha_a / c_a$ represents the sampling rate of action a per unit of cost, where $\alpha_a = c_a \gamma_a \in [0,1]$ is the fraction of the total cost budget allocated to action $a \in \{1, 2, 12\}$.

Proof. Fix any alternative environment $\lambda = (\mu_1', \mu_2', \rho') \in \Theta$ with $\mu_1' < \mu_2'$. Under θ, Arm 1 is optimal ($\mu_1 > \mu_2$), whereas under λ, Arm 2 is optimal ($\mu_2' > \mu_1'$). Consider the $\mathcal{F}_\tau$-measurable event $E = \{\tau < \infty, \widehat{a}_\tau = 1\}$. By δ-PAC correctness under θ and λ, we have $\mathbb{P}_\theta(E) \ge 1 - \delta$ and $\mathbb{P}_\lambda(E) \le \delta$.

By the data-processing inequality for KL divergence applied to the indicator of E and Wald's identity applied to the stopped log-likelihood ratio $\ell_\tau(\theta) - \ell_\tau(\lambda)$ (which is integrable since $\mathbb{E}_\theta[\tau] \le \mathbb{E}_\theta[\mathrm{Cost}(\tau)]/c_{\min} < \infty$), we obtain

$$\begin{aligned} &\mathbb{E}_\theta[N_1(\tau)] D_1(\theta, \lambda) + \mathbb{E}_\theta[N_2(\tau)] D_2(\theta, \lambda) + \mathbb{E}_\theta[N_{1,2}(\tau)] D_{12}(\theta, \lambda) \\ &\quad = \mathbb{E}_\theta[\ell_\tau(\theta) - \ell_\tau(\lambda)] \\ &\quad \ge \mathrm{kl}(\mathbb{P}_\theta(E), \mathbb{P}_\lambda(E)) \ge \mathrm{kl}(1-\delta, \delta) = \mathrm{kl}(\delta, 1-\delta), \end{aligned}$$

where the second inequality uses the monotonicity of $\mathrm{kl}(p,q)$ for $p \ge 1 - \delta > \delta \ge q$ (assuming $\delta < 1/2$). Define the cost-normalized expected allocation weights $\gamma_a := \mathbb{E}_\theta[N_a(\tau)] / \mathbb{E}_\theta[\mathrm{Cost}(\tau)]$ for $a \in \{1, 2, 12\}$. By linearity of expectation in (1), $c_1\gamma_1 + c_2\gamma_2 + c_{1,2}\gamma_{1,2} = 1$. Factoring out $\mathbb{E}_\theta[\mathrm{Cost}(\tau)]$ gives

$$\mathbb{E}_\theta[\mathrm{Cost}(\tau)] \Big[\gamma_1 D_1(\theta, \lambda) + \gamma_2 D_2(\theta, \lambda) + \gamma_{1,2} D_{12}(\theta, \lambda) \Big] \ge \mathrm{kl}(\delta, 1-\delta).$$

Taking the infimum over λ with $\mu_1' < \mu_2'$ (which equals the infimum over its closure Alt_1 by continuity of $D_a(\theta, \cdot)$) and then taking the supremum over all valid γ yields $\mathbb{E}_\theta[\mathrm{Cost}(\tau)] I^\star(\theta) \ge \mathrm{kl}(\delta, 1-\delta)$. Finally, since $\lim_{\delta \downarrow 0} \frac{\mathrm{kl}(\delta, 1-\delta)}{\log(1/\delta)} = 1$, dividing by $\log(1/\delta)$ and taking $\liminf_{\delta \downarrow 0}$ establishes (4). □

3 Analytical Resolution of the Minimax Problem and Phase Transitions

Throughout this section, to lighten notation, we write $\delta := \mu_1 - \mu_2 > 0$ for the suboptimality gap and $\Delta_1 := \mu_1' - \mu_1$, $\Delta_2 := \mu_2' - \mu_2$ for the mean shifts. The alternative condition $\mu_2' \ge \mu_1'$ translates to

$$(\mu_2 + \Delta_2) - (\mu_1 + \Delta_1) \ge 0 \iff \Delta_2 - \Delta_1 \ge \mu_1 - \mu_2 = \delta.$$

Because the quadratic forms in (Δ_1, Δ_2) in (2) and (3) are invariant under simultaneous sign inversion $(\Delta_1, \Delta_2) \mapsto (-\Delta_1, -\Delta_2)$, minimizing over $\Delta_2 - \Delta_1 \ge \delta$ is identical to minimizing over $\Delta_1 - \Delta_2 \ge \delta > 0$.

3.1 Case I: Known Correlation ($\rho' = \rho$)

We first solve the baseline setting where the correlation ρ is known to both the learner and the adversary ($\rho' = \rho$). From (2) and (3) with $\rho' = \rho$, the objective function to be minimized over $\Delta_1 - \Delta_2 \geq \delta$ and maximized over $\gamma = (\gamma_1, \gamma_2, \gamma_{1,2})$ subject to $c_1\gamma_1 + c_2\gamma_2 + c_{1,2}\gamma_{1,2} = 1$ is

$$f(\Delta_1, \Delta_2, \gamma) = \frac{1}{2}\left[\gamma_1\Delta_1^2 + \gamma_2\Delta_2^2 + \gamma_{1,2}\left(\frac{\Delta_1^2 - 2\rho\Delta_1\Delta_2 + \Delta_2^2}{1-\rho^2}\right)\right]. \tag{6}$$

3.1.1 Inner Infimum over (Δ_1, Δ_2)

Grouping the coefficients of Δ_1^2, Δ_2^2, and $\Delta_1\Delta_2$ inside the bracket of (6), we define

$$A := \gamma_1 + \frac{\gamma_{1,2}}{1-\rho^2}, \qquad B := \gamma_2 + \frac{\gamma_{1,2}}{1-\rho^2}, \qquad C := \frac{\rho\gamma_{1,2}}{1-\rho^2},$$

so that $f(\Delta_1, \Delta_2, \gamma) = \frac{1}{2}\big(A\Delta_1^2 + B\Delta_2^2 - 2C\Delta_1\Delta_2\big)$. The Hessian matrix of $A\Delta_1^2 + B\Delta_2^2 - 2C\Delta_1\Delta_2$ with respect to (Δ_1, Δ_2) is $2\begin{pmatrix} A & -C \\ -C & B \end{pmatrix}$, which has trace $2(A+B) > 0$ and determinant

$$4(AB - C^2) = 4\left[\gamma_1\gamma_2 + \frac{(\gamma_1+\gamma_2)\gamma_{1,2}}{1-\rho^2} + \frac{\gamma_{1,2}^2(1-\rho^2)}{(1-\rho^2)^2}\right] = 4\left[\gamma_1\gamma_2 + \frac{(\gamma_1+\gamma_2)\gamma_{1,2} + \gamma_{1,2}^2}{1-\rho^2}\right] > 0$$

whenever at least two actions have positive weight or $\gamma_{1,2} > 0$. Since the unconstrained global minimum is $(0,0) \notin \{\Delta_1 - \Delta_2 \geq \delta\}$, by strict convexity the constrained minimum must lie on the boundary line $\Delta_1 = \Delta_2 + \delta$.

Substituting $\Delta_1 = \Delta_2 + \delta$ into $A\Delta_1^2 + B\Delta_2^2 - 2C\Delta_1\Delta_2$ and expanding gives the 1D quadratic in Δ_2:

$$\begin{aligned} A(\Delta_2+\delta)^2 + B\Delta_2^2 - 2C(\Delta_2+\delta)\Delta_2 &= A(\Delta_2^2 + 2\delta\Delta_2 + \delta^2) + B\Delta_2^2 - 2C(\Delta_2^2 + \delta\Delta_2) \\ &= (A + B - 2C)\Delta_2^2 + 2\delta(A-C)\Delta_2 + A\delta^2. \end{aligned}$$

Setting the derivative with respect to Δ_2 to zero yields

$$2(A+B-2C)\Delta_2 + 2\delta(A-C) = 0 \implies \Delta_2^\star = -\frac{\delta(A-C)}{A+B-2C}.$$

Substituting $\Delta_2^\star$ back into the quadratic gives the inner minimum:

$$\begin{aligned} g(\gamma) := \min_{\Delta_1 - \Delta_2 \geq \delta} f(\Delta_1, \Delta_2, \gamma) &= \frac{\delta^2}{2}\left(A - \frac{(A-C)^2}{A+B-2C}\right) \\ &= \frac{\delta^2}{2}\left(\frac{A^2 + AB - 2AC - (A^2 - 2AC + C^2)}{A+B-2C}\right) \\ &= \frac{\delta^2}{2}\frac{AB - C^2}{A+B-2C}. \end{aligned} \tag{7}$$

Substituting $AB - C^2 = \frac{\gamma_1\gamma_2(1-\rho^2) + (\gamma_1+\gamma_2)\gamma_{1,2} + \gamma_{1,2}^2}{1-\rho^2}$ and

$$A + B - 2C = \gamma_1 + \gamma_2 + \frac{2\gamma_{1,2}(1-\rho)}{1-\rho^2} = \gamma_1 + \gamma_2 + \frac{2\gamma_{1,2}}{1+\rho} = \frac{(\gamma_1+\gamma_2)(1+\rho) + 2\gamma_{1,2}}{1+\rho},$$

and using $\frac{1+\rho}{1-\rho^2} = \frac{1}{1-\rho}$ yields the explicit Case I value function:

$$g(\gamma) = \frac{\delta^2}{2}\left[\frac{(\gamma_1+\gamma_2)\gamma_{1,2} + \gamma_{1,2}^2 + \gamma_1\gamma_2(1-\rho^2)}{(1-\rho)\big[(\gamma_1+\gamma_2)(1+\rho) + 2\gamma_{1,2}\big]}\right]. \tag{8}$$

3.1.2 Outer Supremum over γ

Because $f(\Delta_1, \Delta_2, \gamma)$ is linear in γ for every fixed (Δ_1, Δ_2), its pointwise infimum $g(\gamma)$ is concave on the cost simplex $c_1\gamma_1 + c_2\gamma_2 + c_{1,2}\gamma_{1,2} = 1$. Differentiating $\frac{AB-C^2}{A+B-2C}$ with respect to γ_1, γ_2, and $\gamma_{1,2}$ (using $\frac{\partial A}{\partial \gamma_1} = 1$, $\frac{\partial B}{\partial \gamma_2} = 1$, $\frac{\partial A}{\partial \gamma_{1,2}} = \frac{\partial B}{\partial \gamma_{1,2}} = \frac{1}{1-\rho^2}$, and $\frac{\partial C}{\partial \gamma_{1,2}} = \frac{\rho}{1-\rho^2}$) gives:

$$\begin{aligned}
\frac{\partial}{\partial \gamma_1}\left(\frac{AB-C^2}{A+B-2C}\right) &= \frac{B(A+B-2C)-(AB-C^2)}{(A+B-2C)^2} = \left(\frac{B-C}{A+B-2C}\right)^2, \\
\frac{\partial}{\partial \gamma_2}\left(\frac{AB-C^2}{A+B-2C}\right) &= \frac{A(A+B-2C)-(AB-C^2)}{(A+B-2C)^2} = \left(\frac{A-C}{A+B-2C}\right)^2, \\
\frac{\partial}{\partial \gamma_{1,2}}\left(\frac{AB-C^2}{A+B-2C}\right) &= \frac{\frac{1}{1-\rho^2}(A+B-2\rho C)(A+B-2C)-(AB-C^2)\frac{2}{1+\rho}}{(A+B-2C)^2} \\
&= \frac{(A-C)^2+(B-C)^2+2\rho(A-C)(B-C)}{(1-\rho^2)(A+B-2C)^2}.
\end{aligned}$$

Let $P := (1+\rho)(A-C) = \gamma_1(1+\rho) + \gamma_{1,2}$ and $Q := (1+\rho)(B-C) = \gamma_2(1+\rho) + \gamma_{1,2}$. By the method of Lagrange multipliers on the simplex $c_1\gamma_1 + c_2\gamma_2 + c_{1,2}\gamma_{1,2} = 1$, an interior stationary point requires $\frac{1}{c_1}\frac{\partial g}{\partial \gamma_1} = \frac{1}{c_2}\frac{\partial g}{\partial \gamma_2} = \frac{1}{c_{1,2}}\frac{\partial g}{\partial \gamma_{1,2}}$. From $\frac{1}{c_1}\frac{\partial g}{\partial \gamma_1} = \frac{1}{c_2}\frac{\partial g}{\partial \gamma_2}$, we obtain the exact symmetry relation:

$$\frac{Q^2}{c_1} = \frac{P^2}{c_2} \implies \sqrt{c_2}\big(\gamma_2(1+\rho)+\gamma_{1,2}\big) = \sqrt{c_1}\big(\gamma_1(1+\rho)+\gamma_{1,2}\big) \implies Q = sP, \quad s := \sqrt{\frac{c_1}{c_2}}. \tag{9}$$

Substituting $Q = sP$ into the third condition $\frac{1}{c_{1,2}}\frac{\partial g}{\partial \gamma_{1,2}} = \frac{1}{c_2}\frac{\partial g}{\partial \gamma_2}$ gives:

$$\frac{1}{c_{1,2}}\frac{P^2(1+s^2+2s\rho)}{(1-\rho^2)(1+\rho)^2(A+B-2C)^2} = \frac{1}{c_2}\frac{P^2}{(1+\rho)^2(A+B-2C)^2}.$$

Because $P > 0$ at any interior point, canceling $\frac{P^2}{(1+\rho)^2(A+B-2C)^2}$ shows that an interior stationary point can exist if and only if

$$K_{\mathrm{I}} := c_{1,2}(1-\rho) - c_2\frac{s^2+2s\rho+1}{1+\rho} = 0. \tag{10}$$

Thus, for generic costs where $K_{\mathrm{I}} \neq 0$, the supremum must lie on the boundary of the simplex:

- **Boundary Edge 1 (Pure Joint Sampling, $\gamma_1 = \gamma_2 = 0$, $\gamma_{1,2} = 1/c_{1,2}$):** Evaluating (8) gives $g(0,0,1/c_{1,2}) = \frac{\delta^2}{2}\frac{\gamma_{1,2}^2}{(1-\rho)(2\gamma_{1,2})} = \frac{\delta^2}{4c_{1,2}(1-\rho)}$.
- **Boundary Edge 2 (Pure Individual Sampling, $\gamma_{1,2} = 0$):** Setting $\gamma_{1,2} = 0$ in (9) gives $\sqrt{c_2}\gamma_2 = \sqrt{c_1}\gamma_1$. Combined with $c_1\gamma_1 + c_2\gamma_2 = 1$, this yields $\gamma_1 = \frac{1}{\sqrt{c_1}(\sqrt{c_1}+\sqrt{c_2})}$ and $\gamma_2 = \frac{1}{\sqrt{c_2}(\sqrt{c_1}+\sqrt{c_2})}$, with value $g(\gamma_1,\gamma_2,0) = \frac{\delta^2}{2(\sqrt{c_1}+\sqrt{c_2})^2}$.

Under symmetric individual costs ($c_1 = c_2 = c \implies s = 1$), both the condition $K_{\mathrm{I}} = c_{1,2}(1-\rho) - 2c = 0$ and the boundary crossover $\frac{\delta^2}{4c_{1,2}(1-\rho)} = \frac{\delta^2}{8c}$ yield the exact Case I phase-transition threshold:

$$c_{1,2}^{\mathrm{th,I}}(\rho) = \frac{2c}{1-\rho}. \tag{11}$$

3.2 Case II: Unknown Alternative Correlation ($\rho' \in (-1,1)$)

We now solve the full minimax problem (5) where $\rho' \in (-1,1)$ is also chosen by the adversary to minimize the KL divergence, under symmetric individual pull costs $c_1 = c_2 = c$.

3.2.1 Inner Minimization over (Δ_1, Δ_2) at Fixed ρ'

For any fixed $\rho' \in (-1, 1)$, minimizing $\gamma_1 D_1(\theta, \lambda) + \gamma_2 D_2(\theta, \lambda) + \gamma_{1,2} D_{12}(\theta, \lambda)$ over $\Delta_1 - \Delta_2 \geq \delta$ has the exact same quadratic mean structure as Case I with ρ replaced by ρ', plus the ρ'-only covariance penalty from (3):

$$J(\gamma, \rho') := \frac{\delta^2}{2} g_0(\gamma, \rho') + \frac{\gamma_{1,2}}{2} \left[\log\left(\frac{1 - \rho'^2}{1 - \rho^2} \right) + \frac{2(\rho'^2 - \rho\rho')}{1 - \rho'^2} \right], \tag{12}$$

where

$$g_0(\gamma, \rho') := \frac{\gamma_1\gamma_2(1 - \rho'^2) + \gamma_{1,2}(\gamma_1 + \gamma_2 + \gamma_{1,2})}{(1 - \rho')\big[(\gamma_1 + \gamma_2)(1 + \rho') + 2\gamma_{1,2}\big]}. \tag{13}$$

3.2.2 Rigorous Proof of Symmetric Allocations ($\gamma_1 = \gamma_2$)

Write $S := \gamma_1 + \gamma_2$ for the total individual weight and $D := \gamma_1 - \gamma_2$ for the asymmetry, so $\gamma_1 = \frac{S+D}{2}$, $\gamma_2 = \frac{S-D}{2}$, and $\gamma_1\gamma_2 = \frac{S^2 - D^2}{4}$. The budget constraint $cS + c_{1,2}\gamma_{1,2} = 1$ depends only on S and $\gamma_{1,2}$, and is independent of D. Substituting $\gamma_1\gamma_2 = \frac{S^2-D^2}{4}$ into (13) separates g_0 into a symmetric component and an asymmetry penalty:

$$g_0(\gamma, \rho') = \frac{\frac{S^2}{4}(1 - \rho'^2) + \gamma_{1,2}(S + \gamma_{1,2})}{(1 - \rho')\big[S(1 + \rho') + 2\gamma_{1,2}\big]} - \frac{D^2(1 - \rho'^2)}{4(1 - \rho')\big[S(1 + \rho') + 2\gamma_{1,2}\big]}.$$

Because $1 - \rho'^2 > 0$, $1 - \rho' > 0$, and $1 + \rho' > 0$ for all $\rho' \in (-1, 1)$, the second term is strictly negative whenever $D \neq 0$. Hence, for every fixed S and $\gamma_{1,2}$ and every $\rho' \in (-1, 1)$ simultaneously, $J(\gamma, \rho')$ is strictly maximized at $D = 0$. Taking the infimum over $\rho' \in (-1, 1)$ preserves this dominance, proving that $\gamma_1 = \gamma_2 =: \gamma$ at optimality.

3.2.3 Exact Factorization and Linearization of the Objective

Setting $\gamma_1 = \gamma_2 = \gamma$ in (13) factors the numerator and denominator by the common strictly positive factor $\big[\gamma(1 + \rho') + \gamma_{1,2}\big] > 0$:

$$\begin{aligned} g_0(\gamma, \gamma, \gamma_{1,2}, \rho') &= \frac{\gamma^2(1 - \rho')(1 + \rho') + 2\gamma\gamma_{1,2} + \gamma_{1,2}^2}{(1 - \rho')\big[2\gamma(1 + \rho') + 2\gamma_{1,2}\big]} \\ &= \frac{\big[\gamma(1 - \rho') + \gamma_{1,2}\big]\cancel{\big[\gamma(1 + \rho') + \gamma_{1,2}\big]}}{2(1 - \rho')\cancel{\big[\gamma(1 + \rho') + \gamma_{1,2}\big]}} = \frac{\gamma}{2} + \frac{\gamma_{1,2}}{2(1 - \rho')}. \end{aligned} \tag{14}$$

Substituting (14) into (12) makes $J(\gamma, \rho')$ strictly affine in $(\gamma, \gamma_{1,2})$:

$$J(\gamma, \rho') = \frac{\delta^2}{4}\gamma + \frac{\gamma_{1,2}}{2} \underbrace{\left[\frac{\delta^2}{2(1 - \rho')} + \log\left(\frac{1 - \rho'^2}{1 - \rho^2} \right) + \frac{2(\rho'^2 - \rho\rho')}{1 - \rho'^2} \right]}_{=: f(\rho')}. \tag{15}$$

3.2.4 Worst-Case Correlation $\rho'^\star$ and Cost Threshold $c_{1,2}^{\text{th}}$

Since $\frac{\delta^2}{4}\gamma$ is independent of ρ' and $\gamma_{1,2} \geq 0$, taking the infimum of (15) over $\rho' \in (-1, 1)$ reduces to computing $K(\delta, \rho) := \inf_{\rho' \in (-1,1)} f(\rho')$. Differentiating each term of $f(\rho')$ with respect to ρ' over the common denominator $2(1 - \rho'^2)^2$ gives:

$$\begin{aligned} \frac{d}{d\rho'}\left[\frac{\delta^2}{2(1 - \rho')} \right] &= \frac{\delta^2}{2(1 - \rho')^2} = \frac{\delta^2(1 + \rho')^2}{2(1 - \rho'^2)^2}, \\ \frac{d}{d\rho'}\big[\log(1 - \rho'^2)\big] &= -\frac{2\rho'}{1 - \rho'^2} = \frac{-4\rho' + 4\rho'^3}{2(1 - \rho'^2)^2}, \end{aligned}$$

$$\frac{d}{d\rho'}\left[\frac{2\rho'^2-2\rho\rho'}{1-\rho'^2}\right]=\frac{(4\rho'-2\rho)(1-\rho'^2)+2\rho'(2\rho'^2-2\rho\rho')}{(1-\rho'^2)^2}=\frac{8\rho'-4\rho-4\rho\rho'^2}{2(1-\rho'^2)^2}.$$

Summing the numerators gives

$$f'(\rho')=\frac{\delta^2(1+\rho')^2+4\rho'^3-4\rho\rho'^2+4\rho'-4\rho}{2(1-\rho'^2)^2}=\frac{\delta^2(1+\rho')^2+4(\rho'^2+1)(\rho'-\rho)}{2(1-\rho'^2)^2}.$$

Thus, stationary points $\rho'^\star\in(-1,1)$ satisfy the scalar cubic equation

$$C(\rho'):=\delta^2(1+\rho')^2+4\rho'^3-4\rho\rho'^2+4\rho'-4\rho=\delta^2(1+\rho')^2+4(\rho'^2+1)(\rho'-\rho)=0. \tag{16}$$

Evaluating $C(\rho')$ at $\rho'=-1$ and $\rho'=\rho$ gives $C(-1)=-8(1+\rho)<0$ and $C(\rho)=\delta^2(1+\rho)^2>0$ (and $C(\rho')>0$ for all $\rho'\in[\rho,1]$). By the Intermediate Value Theorem, there exists a root $\rho'^\star\in(-1,\rho)$ solving $C(\rho'^\star)=0$. Substituting $K=f(\rho'^\star)$ and $\gamma_{1,2}=\frac{1-2c\gamma}{c_{1,2}}$ into (15) establishes the following theorem:

Theorem 3.1 (Vertex Optimality and Phase Transition)**.** *Under symmetric individual pull costs $c_1=c_2=c$, the minimax objective on the cost simplex $2c\gamma+c_{1,2}\gamma_{1,2}=1$ reduces to maximizing a linear function of $\gamma\in[0,1/(2c)]$:*

$$I^\star(\theta)=\sup_{\gamma\in[0,\frac{1}{2c}]}\left[\gamma\left(\frac{\delta^2}{4}-\frac{cK}{c_{1,2}}\right)+\frac{K}{2c_{1,2}}\right]=\max\Big\{I(\theta;v_I),\,I(\theta;v_J)\Big\},$$

where $v_I=\left(\frac{1}{2c},\frac{1}{2c},0\right)$ with $I(\theta;v_I)=\frac{\delta^2}{8c}$, and $v_J=\left(0,0,\frac{1}{c_{1,2}}\right)$ with $I(\theta;v_J)=\frac{K}{2c_{1,2}}$. The phase transition between pure joint sampling and symmetric individual sampling occurs at the cost threshold:

$$c^{\text{th}}_{1,2}(\delta,\rho)=\frac{4cK}{\delta^2}=\frac{4c}{\delta^2}\left[\frac{\delta^2}{2(1-\rho'^\star)}+\log\left(\frac{1-(\rho'^\star)^2}{1-\rho^2}\right)+\frac{2\rho'^\star(\rho'^\star-\rho)}{1-(\rho'^\star)^2}\right]. \tag{17}$$

- *If $c_{1,2}<c^{\text{th}}_{1,2}(\delta,\rho)$, pure joint sampling ($v_J$) is uniquely optimal.*
- *If $c_{1,2}>c^{\text{th}}_{1,2}(\delta,\rho)$, symmetric individual sampling (v_I) is uniquely optimal.*
- ***Four Asymptotic Limits (with $c=1$):***
  - *(i)* Large Gap ($\delta\to\infty$): *Dividing $C(\rho')=0$ by δ^2 gives $(1+\rho')^2\to0\implies\rho'^\star\to-1$, and $c^{\text{th}}_{1,2}\to\frac{4}{\delta^2}\left[\frac{\delta^2}{2(1-(-1))}\right]=1$.*
  - *(ii)* Vanishing Gap ($\delta\to0$): *Setting $\delta\to0$ in $C(\rho')=0$ gives $4(\rho'^2+1)(\rho'-\rho)=0\implies\rho'^\star\to\rho$, and $c^{\text{th}}_{1,2}\to\frac{4}{\delta^2}\left[\frac{\delta^2}{2(1-\rho)}\right]=\frac{2}{1-\rho}$, recovering the Case I threshold (11).*
  - *(iii)* Perfect Positive Correlation ($\rho\to1$): *Writing $\rho'=1-\epsilon$ in $C(\rho')=0$ at $\rho=1$ gives $\delta^2(2-\epsilon)^2-4\epsilon(2-2\epsilon+\epsilon^2)=0\implies4\delta^2-8\epsilon\approx0\implies\rho'^\star\to1-\frac{\delta^2}{2}$, and $c^{\text{th}}_{1,2}\to\infty$.*
  - *(iv)* Perfect Negative Correlation ($\rho\to-1$): *As $\rho\to-1$, $C(\rho')\to(1+\rho')\left[\delta^2(1+\rho')+4(\rho'^2+1)\right]$, whose only root in $[-1,1)$ is $\rho'=-1$, so $\rho'^\star\to-1$, and $c^{\text{th}}_{1,2}\to1$.*

Figure 1 shows $c^{\text{th}}_{1,2}$ along slices of fixed ρ and fixed δ, and the limits (i)–(iii) of Theorem 3.1 can be read off the curves.

Figure 2 illustrates Theorem 3.1 for a fixed θ: the optimal allocation stays at v_J until $c_{1,2}=c^{\text{th}}_{1,2}$ and then switches to v_I.

Remark 3.2 (Case I vs. Case II)**.** In Case I (known ρ), the adversary is restricted to $\rho'=\rho$, where $f(\rho)=\frac{\delta^2}{2(1-\rho)}$. In Case II (unknown ρ), the adversary minimizes $f(\rho')$ over $\rho'\in(-1,1)$. Because $f'(\rho)=\frac{\delta^2}{2(1-\rho)^2}>0$ for every $\delta>0$, the unconstrained minimizer satisfies $\rho'^\star<\rho$

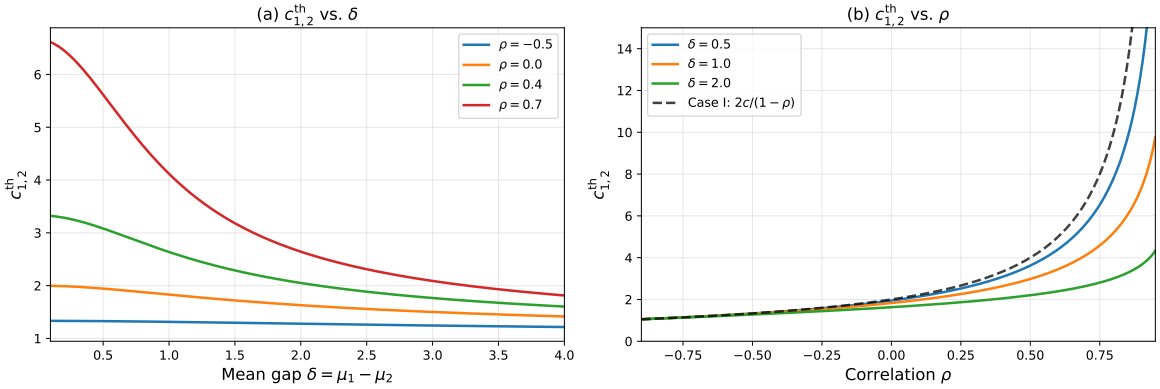


Figure 1: $c^{\text{th}}_{1,2}(\delta, \rho)$ from (17) with $c = 1$. (a) As a function of δ for fixed ρ; each curve starts at $2/(1-\rho)$ as $\delta \to 0$. (b) As a function of ρ for fixed δ, together with the Case I threshold $2/(1-\rho)$ (dashed).

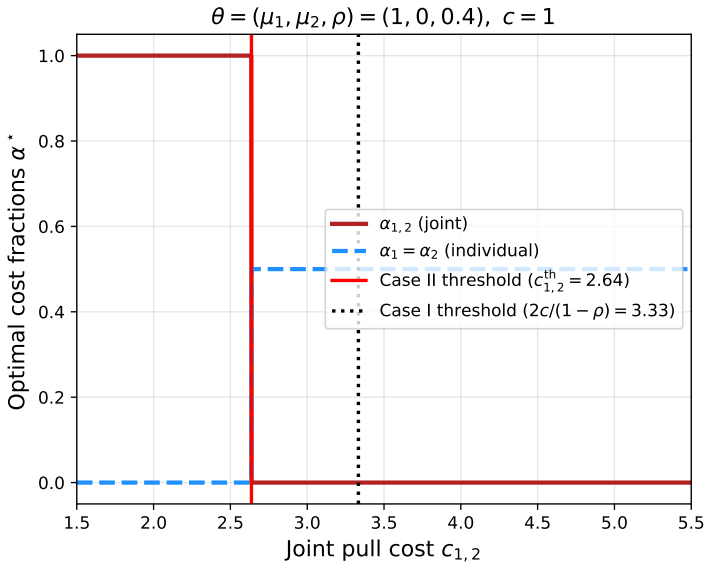
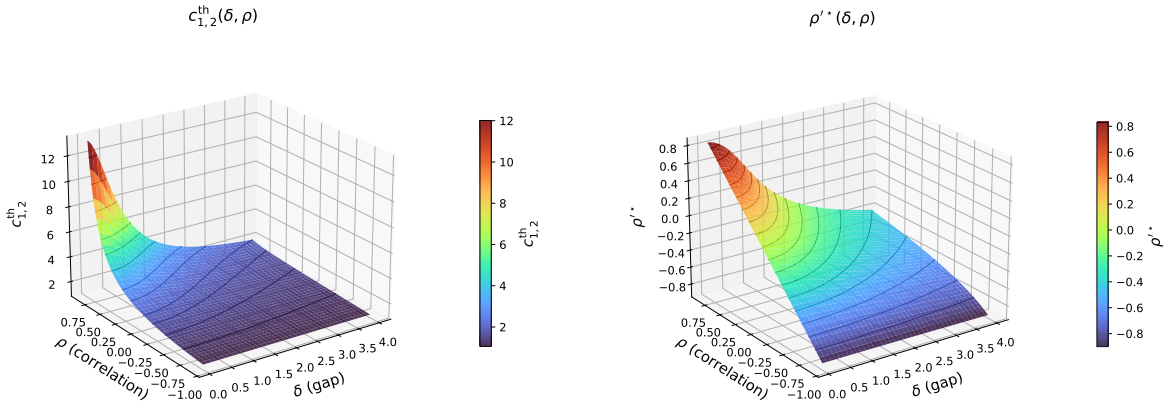


Figure 2: Optimal cost fractions $\alpha_a = c_a \gamma_a$ as a function of $c_{1,2}$ for $\theta = (1, 0, 0.4)$ and $c = 1$. The allocation switches from v_J to v_I at $c^{\text{th}}_{1,2}(\delta, \rho) = 2.64$ given by (17) (solid red line); the dotted line is the Case I threshold $2c/(1-\rho) = 3.33$ from (11).

and $K(\delta,\rho) = f(\rho'^\star) < f(\rho) = \frac{\delta^2}{2(1-\rho)}$. Consequently, whenever joint pulls are used, unknown correlation strictly reduces $I^\star(\theta)$ ($I^\star_{\text{Case II}}(\theta) \le I^\star_{\text{Case I}}(\theta)$), which strictly increases the asymptotic cost lower bound ($\frac{1}{I^\star_{\text{Case II}}(\theta)} \ge \frac{1}{I^\star_{\text{Case I}}(\theta)}$) and shrinks the phase-transition boundary ($c^{\text{th}}_{1,2}(\delta,\rho) < \frac{2c}{1-\rho}$). As $\delta \to 0$, $\rho'^\star - \rho = -\frac{\delta^2(1+\rho)^2}{4(1+\rho^2)} + O(\delta^4)$, so $f(\rho) - K(\delta,\rho) = O(\delta^4)$ and estimating ρ becomes asymptotically free in the small-gap limit.

Figure 3 shows $c^{\text{th}}_{1,2}$ and $\rho'^\star$ over (δ,ρ); $\rho'^\star$ stays close to ρ for small δ and moves towards -1 as δ grows.



(a) $c^{\text{th}}_{1,2}(\delta,\rho)$ (b) $\rho'^\star(\delta,\rho)$

Figure 3: (a) The threshold $c^{\text{th}}_{1,2}$ from (17) and (b) the minimizing correlation $\rho'^\star$, the root of (16) in $(-1,\rho)$, over $\delta \in (0,4]$ and $\rho \in [-0.9, 0.85]$, with $c = 1$.

4 Exact Derivation of the Profiled GLR Stopping Statistic Z_t

We now derive the closed-form Generalized Likelihood Ratio (GLR) stopping statistic Z_t for the two-arm model. After N_1 pulls of Arm 1, N_2 pulls of Arm 2, and $N_{1,2}$ joint pulls, write the four empirical sample averages as

$$\overline{X}_1 = \frac{1}{N_1}\sum_{i=1}^{N_1} X_{1,i}, \qquad \overline{X}_2 = \frac{1}{N_2}\sum_{i=1}^{N_2} X_{2,i}, \qquad \overline{U} = \frac{1}{N_{1,2}}\sum_{j=1}^{N_{1,2}} U_j, \qquad \overline{V} = \frac{1}{N_{1,2}}\sum_{j=1}^{N_{1,2}} V_j.$$

The joint log-likelihood of all observations up to round t at a candidate parameter $\theta' = (\mu'_1, \mu'_2, \rho') \in \Theta$ is

$$\begin{aligned}\ell_t(\mu'_1,\mu'_2,\rho') = &-\frac{1}{2}\sum_{i=1}^{N_1}(X_{1,i}-\mu'_1)^2 - \frac{1}{2}\sum_{i=1}^{N_2}(X_{2,i}-\mu'_2)^2 - \frac{N_{1,2}}{2}\log(1-\rho'^2)\\ &- \frac{1}{2(1-\rho'^2)}\sum_{j=1}^{N_{1,2}}\Big[(U_j-\mu'_1)^2 - 2\rho'(U_j-\mu'_1)(V_j-\mu'_2) + (V_j-\mu'_2)^2\Big] + \text{const}, \quad (18)\end{aligned}$$

where const $= -\frac{N_1+N_2+2N_{1,2}}{2}\log(2\pi)$ cancels out in all likelihood ratios.

4.1 Unconstrained MLE via Cramer's Rule and 1D Profiling

Differentiating (18) with respect to μ_1', μ_2', and ρ' and setting each derivative to zero gives the three first-order score equations:

$$\frac{\partial \ell_t}{\partial \mu_1'} = N_1(\overline{X}_1 - \mu_1') + \frac{N_{1,2}}{1-\rho'^2}(\overline{U} - \mu_1') - \frac{N_{1,2}\rho'}{1-\rho'^2}(\overline{V} - \mu_2') = 0, \tag{I}$$

$$\frac{\partial \ell_t}{\partial \mu_2'} = N_2(\overline{X}_2 - \mu_2') + \frac{N_{1,2}}{1-\rho'^2}(\overline{V} - \mu_2') - \frac{N_{1,2}\rho'}{1-\rho'^2}(\overline{U} - \mu_1') = 0, \tag{II}$$

$$\frac{\partial \ell_t}{\partial \rho'} = \frac{1}{(1-\rho'^2)^2}\Big[N_{1,2}\rho'(1-\rho'^2) - \rho'\sum_{j=1}^{N_{1,2}}\left[(U_j-\mu_1')^2 + (V_j-\mu_2')^2\right] + (1+\rho'^2)\sum_{j=1}^{N_{1,2}}(U_j-\mu_1')(V_j-\mu_2')\Big] = 0. \tag{III}$$

For any fixed $\rho' \in (-1,1)$, Equations (I) and (II) form a 2×2 linear system in (μ_1', μ_2'):

$$\begin{pmatrix} N_1 + \frac{N_{1,2}}{1-\rho'^2} & -\frac{N_{1,2}\rho'}{1-\rho'^2} \\ -\frac{N_{1,2}\rho'}{1-\rho'^2} & N_2 + \frac{N_{1,2}}{1-\rho'^2} \end{pmatrix}\begin{pmatrix}\mu_1' \\ \mu_2'\end{pmatrix} = \begin{pmatrix} N_1\overline{X}_1 + \frac{N_{1,2}}{1-\rho'^2}\overline{U} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{V} \\ N_2\overline{X}_2 + \frac{N_{1,2}}{1-\rho'^2}\overline{V} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{U}\end{pmatrix},$$

whose determinant is strictly positive for all $\rho' \in (-1,1)$ whenever $N_{1,2} \geq 1$:

$$\begin{aligned} D(\rho') &:= \left(N_1 + \frac{N_{1,2}}{1-\rho'^2}\right)\left(N_2 + \frac{N_{1,2}}{1-\rho'^2}\right) - \left(\frac{N_{1,2}\rho'}{1-\rho'^2}\right)^2 \\ &= N_1N_2 + \frac{(N_1+N_2)N_{1,2} + N_{1,2}^2}{1-\rho'^2} > 0. \end{aligned}$$

Solving via Cramer's rule gives the profiled means $\mu_1'(\rho')$ and $\mu_2'(\rho')$ as explicit rational functions of ρ':

$$\mu_1'(\rho') = \frac{\left(N_2 + \frac{N_{1,2}}{1-\rho'^2}\right)\left(N_1\overline{X}_1 + \frac{N_{1,2}}{1-\rho'^2}\overline{U} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{V}\right) + \frac{N_{1,2}\rho'}{1-\rho'^2}\left(N_2\overline{X}_2 + \frac{N_{1,2}}{1-\rho'^2}\overline{V} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{U}\right)}{D(\rho')}, \tag{19}$$

$$\mu_2'(\rho') = \frac{\frac{N_{1,2}\rho'}{1-\rho'^2}\left(N_1\overline{X}_1 + \frac{N_{1,2}}{1-\rho'^2}\overline{U} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{V}\right) + \left(N_1 + \frac{N_{1,2}}{1-\rho'^2}\right)\left(N_2\overline{X}_2 + \frac{N_{1,2}}{1-\rho'^2}\overline{V} - \frac{N_{1,2}\rho'}{1-\rho'^2}\overline{U}\right)}{D(\rho')}. \tag{20}$$

Substituting $(\mu_1'(\rho'), \mu_2'(\rho'))$ into (III) reduces the 3D optimization to a 1D root-finding problem in $\rho' \in (-1,1)$, yielding $\widehat{\rho}$, $\widehat{\mu}_1 = \mu_1'(\widehat{\rho})$, and $\widehat{\mu}_2 = \mu_2'(\widehat{\rho})$.

4.2 Constrained MLE on the Boundary $\mu_1' = \mu_2' = m$

Suppose $\widehat{\mu}_1 \geq \widehat{\mu}_2$ so that $\widehat{a}_t = 1$ (if $\widehat{\mu}_2 > \widehat{\mu}_1$, swap labels 1 and 2). On $\mathrm{Alt}(1) = \{\mu_2' \geq \mu_1'\}$, because $\ell_t(\mu_1', \mu_2', \rho')$ is strictly concave in (μ_1', μ_2') for each fixed $\rho' \in (-1,1)$, the constrained maximum over $\mu_2' \geq \mu_1'$ lies on the boundary $\mu_1' = \mu_2' = m$. Setting $\mu_1' = \mu_2' = m$ in (18) gives

$$\begin{aligned} \ell_t(m, m, \rho') = &-\frac{1}{2}\sum_{i=1}^{N_1}(X_{1,i}-m)^2 - \frac{1}{2}\sum_{i=1}^{N_2}(X_{2,i}-m)^2 - \frac{N_{1,2}}{2}\log(1-\rho'^2) \\ &- \frac{1}{2(1-\rho'^2)}\sum_{j=1}^{N_{1,2}}\Big[(U_j-m)^2 - 2\rho'(U_j-m)(V_j-m) + (V_j-m)^2\Big]. \end{aligned}$$

Differentiating the joint bracket with respect to m gives $\frac{d}{dm}\big[(U_j-m)^2-2\rho'(U_j-m)(V_j-m)+(V_j-m)^2\big]=-2(1-\rho')\big[(U_j-m)+(V_j-m)\big]$. Using $\frac{1-\rho'}{1-\rho'^2}=\frac{1}{1+\rho'}$, the first-order condition $\frac{\partial}{\partial m}\ell_t(m,m,\rho')=0$ is

$$N_1(\overline{X}_1-m)+N_2(\overline{X}_2-m)+\frac{N_{1,2}}{1+\rho'}\big[(\overline{U}-m)+(\overline{V}-m)\big]=0,$$

which solves directly for the profiled common mean:

$$\widehat{m}(\rho')=\frac{N_1\overline{X}_1+N_2\overline{X}_2+\frac{N_{1,2}(\overline{U}+\overline{V})}{1+\rho'}}{N_1+N_2+\frac{2N_{1,2}}{1+\rho'}}. \tag{21}$$

Solving (III) with $\mu_1'=\mu_2'=\widehat{m}(\rho')$ over $\rho'\in(-1,1)$ gives $\widehat{\rho}'$ and $\widehat{m}=\widehat{m}(\widehat{\rho}')$.

4.3 Step-by-Step Algebraic Subtraction for $Z_t=\ell_t(\widehat{\mu}_1,\widehat{\mu}_2,\widehat{\rho})-\ell_t(\widehat{m},\widehat{m},\widehat{\rho}')$

We now evaluate $Z_t=\ell_t(\widehat{\mu}_1,\widehat{\mu}_2,\widehat{\rho})-\ell_t(\widehat{m},\widehat{m},\widehat{\rho}')$ term by term:

1. **Individual Arm 1 and Arm 2 Contributions:** Writing $X_{1,i}-\widehat{m}=(X_{1,i}-\widehat{\mu}_1)+(\widehat{\mu}_1-\widehat{m})$ and expanding the square gives

$$\begin{aligned}-\frac{1}{2}\sum_{i=1}^{N_1}(X_{1,i}-\widehat{\mu}_1)^2+\frac{1}{2}\sum_{i=1}^{N_1}(X_{1,i}-\widehat{m})^2&=(\widehat{\mu}_1-\widehat{m})\sum_{i=1}^{N_1}(X_{1,i}-\widehat{\mu}_1)+\frac{N_1(\widehat{\mu}_1-\widehat{m})^2}{2}\\&=N_1(\widehat{\mu}_1-\widehat{m})(\overline{X}_1-\widehat{\mu}_1)+\frac{N_1(\widehat{\mu}_1-\widehat{m})^2}{2}.\end{aligned}$$

   From Score Equation (I) evaluated at $(\widehat{\mu}_1,\widehat{\mu}_2,\widehat{\rho})$, we substitute $N_1(\overline{X}_1-\widehat{\mu}_1)=-\frac{N_{1,2}[(\overline{U}-\widehat{\mu}_1)-\widehat{\rho}(\overline{V}-\widehat{\mu}_2)]}{1-\widehat{\rho}^2}$, and similarly for Arm 2 using Score Equation (II): $N_2(\overline{X}_2-\widehat{\mu}_2)=-\frac{N_{1,2}[(\overline{V}-\widehat{\mu}_2)-\widehat{\rho}(\overline{U}-\widehat{\mu}_1)]}{1-\widehat{\rho}^2}$.

2. **Joint Pull Contributions:** Expanding $U_j-\widehat{m}=(U_j-\widehat{\mu}_1)+(\widehat{\mu}_1-\widehat{m})$ and $V_j-\widehat{m}=(V_j-\widehat{\mu}_2)+(\widehat{\mu}_2-\widehat{m})$ inside $\frac{1}{2(1-\widehat{\rho}'^2)}\sum_{j=1}^{N_{1,2}}\big[(U_j-\widehat{m})^2-2\widehat{\rho}'(U_j-\widehat{m})(V_j-\widehat{m})+(V_j-\widehat{m})^2\big]$ separates the sum into three parts:

   - a constant mean-shift quadratic:

$$\frac{N_{1,2}\big[(\widehat{\mu}_1-\widehat{m})^2-2\widehat{\rho}'(\widehat{\mu}_1-\widehat{m})(\widehat{\mu}_2-\widehat{m})+(\widehat{\mu}_2-\widehat{m})^2\big]}{2(1-\widehat{\rho}'^2)};$$

   - linear cross-terms in $(\widehat{\mu}_1-\widehat{m})$ and $(\widehat{\mu}_2-\widehat{m})$:

$$N_{1,2}(\widehat{\mu}_1-\widehat{m})\frac{(\overline{U}-\widehat{\mu}_1)-\widehat{\rho}'(\overline{V}-\widehat{\mu}_2)}{1-\widehat{\rho}'^2}+N_{1,2}(\widehat{\mu}_2-\widehat{m})\frac{(\overline{V}-\widehat{\mu}_2)-\widehat{\rho}'(\overline{U}-\widehat{\mu}_1)}{1-\widehat{\rho}'^2};$$

   - pure residual quadratics in $u_j:=U_j-\widehat{\mu}_1$ and $v_j:=V_j-\widehat{\mu}_2$ evaluated at $\widehat{\rho}'$ and $\widehat{\rho}$. Applying the exact algebraic identity

$$\frac{u_j^2-2ru_jv_j+v_j^2}{2(1-r^2)}=\frac{(u_j+v_j)^2}{4(1+r)}+\frac{(u_j-v_j)^2}{4(1-r)} \tag{22}$$

   at $r=\widehat{\rho}'$ and $r=\widehat{\rho}$, and using $\frac{1}{1+\widehat{\rho}'}-\frac{1}{1+\widehat{\rho}}=\frac{\widehat{\rho}-\widehat{\rho}'}{(1+\widehat{\rho}')(1+\widehat{\rho})}$ and $\frac{1}{1-\widehat{\rho}'}-\frac{1}{1-\widehat{\rho}}=-\frac{\widehat{\rho}-\widehat{\rho}'}{(1-\widehat{\rho}')(1-\widehat{\rho})}$, combines the residual quadratics into sum and difference terms.

Proposition 4.1: Closed-Form Expression for the Stopping Statistic Z_t

Proposition 4.1 (Closed-Form Expression for the Stopping Statistic Z_t)**.** *The GLR stopping statistic $Z_t = \ell_t(\widehat{\mu}_1, \widehat{\mu}_2, \widehat{\rho}) - \ell_t(\widehat{m}, \widehat{m}, \widehat{\rho}')$ is given by*

$$
\begin{aligned}
Z_t = {} & \frac{N_1(\widehat{\mu}_1 - \widehat{m})^2}{2} + \frac{N_2(\widehat{\mu}_2 - \widehat{m})^2}{2} + \frac{N_{1,2}\big[(\widehat{\mu}_1 - \widehat{m})^2 - 2\widehat{\rho}'(\widehat{\mu}_1 - \widehat{m})(\widehat{\mu}_2 - \widehat{m}) + (\widehat{\mu}_2 - \widehat{m})^2\big]}{2(1 - \widehat{\rho}'^2)} \\
& + \frac{N_{1,2}}{2} \log \frac{1 - \widehat{\rho}'^2}{1 - \widehat{\rho}^2} \\
& + \frac{\widehat{\rho} - \widehat{\rho}'}{4} \left[\frac{\sum_{j=1}^{N_{1,2}} \big[(U_j - \widehat{\mu}_1) + (V_j - \widehat{\mu}_2)\big]^2}{(1 + \widehat{\rho}')(1 + \widehat{\rho})} - \frac{\sum_{j=1}^{N_{1,2}} \big[(U_j - \widehat{\mu}_1) - (V_j - \widehat{\mu}_2)\big]^2}{(1 - \widehat{\rho}')(1 - \widehat{\rho})} \right] \\
& + N_{1,2}(\widehat{\mu}_1 - \widehat{m}) \left[\frac{(\overline{U} - \widehat{\mu}_1) - \widehat{\rho}'(\overline{V} - \widehat{\mu}_2)}{1 - \widehat{\rho}'^2} - \frac{(\overline{U} - \widehat{\mu}_1) - \widehat{\rho}(\overline{V} - \widehat{\mu}_2)}{1 - \widehat{\rho}^2} \right] \\
& + N_{1,2}(\widehat{\mu}_2 - \widehat{m}) \left[\frac{(\overline{V} - \widehat{\mu}_2) - \widehat{\rho}'(\overline{U} - \widehat{\mu}_1)}{1 - \widehat{\rho}'^2} - \frac{(\overline{V} - \widehat{\mu}_2) - \widehat{\rho}(\overline{U} - \widehat{\mu}_1)}{1 - \widehat{\rho}^2} \right]. \qquad (23)
\end{aligned}
$$

5 Non-Asymptotic δ-PAC Soundness and the Curvature Bound M_t

5.1 Reference Measure and Mixture Martingale

Because $Y_s \in \mathbb{R}$ when $A_s \in \{1, 2\}$ and $Y_s \in \mathbb{R}^2$ when $A_s = 12$, the dimension of the observation space at round t depends on the random action sequence $(A_1, \ldots, A_t)$. Define the single-round conditional reference measure $\lambda^{(s)}(\cdot \mid A_s)$ to be 1D Lebesgue measure on $\mathbb{R}$ if $A_s \in \{1, 2\}$ and 2D Lebesgue measure on $\mathbb{R}^2$ if $A_s = 12$, and define the t-round recursive reference measure $\lambda_t := \bigotimes_{s=1}^{t} \lambda^{(s)}(\cdot \mid A_s)$.

Lemma 5.1 (L_t Integrates to 1 under λ_t)**.** *For every fixed parameter $\theta' \in \Theta$ and every round $t \geq 0$, $\int L_t(\theta')\, d\lambda_t = 1$.*

Proof. We proceed by induction on t. For $t = 0$, $L_0(\theta') = 1$ and λ_0 is the point mass on the empty history, so $\int L_0\, d\lambda_0 = 1$. Assume the claim holds at round $t - 1$. Factoring $L_t(\theta')(y_{1:t}) = L_{t-1}(\theta')(y_{1:t-1}) p_{\theta'}(y_t \mid A_t(y_{1:t-1}))$ and holding $y_{1:t-1}$ fixed, the action A_t is $\mathcal{F}_{t-1}$-measurable and hence deterministic. Thus the inner integral $\int p_{\theta'}(y_t \mid A_t)\, d\lambda^{(t)}(y_t \mid A_t) = 1$ whether $A_t \in \{1, 2\}$ or $A_t = 12$. Integrating $L_{t-1}(\theta')(y_{1:t-1}) \cdot 1$ over $d\lambda_{t-1}$ yields 1 by the induction hypothesis. □

For any prior probability measure $\mathcal{P}$ on Θ with positive continuous density $p(\theta') > 0$, decompose $Z_t = W_t + \Delta_t$, where

$$
W_t := \log \left(\frac{\int_\Theta L_t(\theta')\, d\mathcal{P}(\theta')}{\sup_{\theta' \in \mathrm{Alt}(\widehat{a}_t)} L_t(\theta')} \right), \qquad \Delta_t := \log \left(\frac{\sup_{\theta' \in \Theta} L_t(\theta')}{\int_\Theta L_t(\theta')\, d\mathcal{P}(\theta')} \right) \geq 0. \qquad (24)
$$

Theorem 5.2 (Non-Asymptotic δ-PAC Soundness of W_t and Z_t)**.** *Fix any arm $i \in \{1, 2\}$ and any true parameter $\theta^\star \in \mathrm{Alt}(i)$. If the stopping threshold satisfies $\beta(t, \delta) \geq \log(1/\delta) + \Delta_t$ pathwise for all $t \geq 1$, then under any adapted sampling rule, the stopping time $\tau_\delta = \inf\{t \geq 1 : Z_t > \beta(t, \delta)\}$ satisfies $\mathbb{P}_{\theta^\star}(\tau_\delta < \infty, \widehat{a}_{\tau_\delta} = i) \leq \delta$.*

Proof. Define the pairwise disjoint stopping events $E_i^t := \{y_{1:t} : \tau_\delta(y_{1:t}) = t, \widehat{a}_{\tau_\delta}(y_{1:t}) = i\}$ for $t \geq 1$. Since $\theta^\star \in \mathrm{Alt}(i)$, we have $L_t(\theta^\star) \leq \sup_{\theta' \in \mathrm{Alt}(i)} L_t(\theta')$. On E_i^t, $\widehat{a}_t = i$ and $Z_t > \beta(t, \delta) \geq \log(1/\delta) + \Delta_t$, which implies $W_t = Z_t - \Delta_t > \log(1/\delta)$, or equivalently $\sup_{\theta' \in \mathrm{Alt}(i)} L_t(\theta') < \delta \int_\Theta L_t(\theta')\, d\mathcal{P}(\theta')$.

Summing over $t \geq 1$ and applying Tonelli's theorem [6] and Lemma 5.1:

$$\begin{aligned}
\mathbb{P}_{\theta^\star}(\tau_\delta < \infty,\, \widehat{a}_{\tau_\delta} = i) &= \sum_{t=1}^{\infty} \int_{E_i^t} L_t(\theta^\star)\, d\lambda_t \\
&\leq \sum_{t=1}^{\infty} \int_{E_i^t} \delta \left(\int_\Theta L_t(\theta')\, d\mathcal{P}(\theta') \right) d\lambda_t \\
&= \delta \int_\Theta \left(\sum_{t=1}^{\infty} \int_{E_i^t} L_t(\theta')\, d\lambda_t \right) d\mathcal{P}(\theta') \\
&= \delta \int_\Theta \mathbb{P}_{\theta'}(\tau_\delta < \infty,\, \widehat{a}_{\tau_\delta} = i)\, d\mathcal{P}(\theta') \leq \delta \int_\Theta 1\, d\mathcal{P}(\theta') = \delta.
\end{aligned}$$

Equivalently, $M_t^{\mathcal{P}} := \frac{\int_\Theta L_t(\theta')\, d\mathcal{P}(\theta')}{L_t(\theta^\star)}$ is a non-negative $\mathcal{F}_t$-martingale with $\mathbb{E}_{\theta^\star}[M_0^{\mathcal{P}}] = 1$, and Ville's maximal inequality [4] gives $\mathbb{P}_{\theta^\star}(\exists t \geq 1 : M_t^{\mathcal{P}} > 1/\delta) \leq \delta$. □

5.2 Analytical Negative Hessian and Definition of M_t

Differentiating the score equations (I)–(III) yields the entries $H_{ab} = -\frac{\partial^2 \ell_t}{\partial \theta'_a \partial \theta'_b}$ of the negative Hessian matrix $H_t(\theta')$:

$$H_{11} = N_1 + \frac{N_{1,2}}{1-\rho'^2}, \qquad H_{22} = N_2 + \frac{N_{1,2}}{1-\rho'^2}, \qquad H_{12} = H_{21} = -\frac{N_{1,2}\rho'}{1-\rho'^2}, \tag{25}$$

$$H_{13} = H_{31} = -\frac{2\rho' N_{1,2}}{(1-\rho'^2)^2}(\overline{U} - \mu'_1) + \frac{(1+\rho'^2)N_{1,2}}{(1-\rho'^2)^2}(\overline{V} - \mu'_2), \tag{26}$$

$$H_{23} = H_{32} = -\frac{2\rho' N_{1,2}}{(1-\rho'^2)^2}(\overline{V} - \mu'_2) + \frac{(1+\rho'^2)N_{1,2}}{(1-\rho'^2)^2}(\overline{U} - \mu'_1), \tag{27}$$

$$\begin{aligned}
H_{33} = &-\frac{N_{1,2}(1+\rho'^2)}{(1-\rho'^2)^2} + \frac{1+3\rho'^2}{(1-\rho'^2)^3} \sum_{j=1}^{N_{1,2}} \left[(U_j - \mu'_1)^2 + (V_j - \mu'_2)^2 \right] \\
&- \frac{2\rho'(3+\rho'^2)}{(1-\rho'^2)^3} \sum_{j=1}^{N_{1,2}} (U_j - \mu'_1)(V_j - \mu'_2).
\end{aligned} \tag{28}$$

Define the following empirical quantities at round t from the observed data and the unconstrained MLE $\widehat{\theta} = (\widehat{\mu}_1, \widehat{\mu}_2, \widehat{\rho})$:

$$\begin{aligned}
\epsilon_0 &:= \frac{1-\widehat{\rho}^2}{2} \in \left(0, \frac{1}{2}\right], \qquad \widehat{\Delta}_{\text{mean}} := \max\left(|\overline{U} - \widehat{\mu}_1|,\, |\overline{V} - \widehat{\mu}_2|\right) + 1 \geq 1, \\
\widehat{\sigma}_U^2 &:= \frac{1}{N_{1,2}} \sum_{j=1}^{N_{1,2}} (U_j - \widehat{\mu}_1)^2, \qquad \widehat{\sigma}_V^2 := \frac{1}{N_{1,2}} \sum_{j=1}^{N_{1,2}} (V_j - \widehat{\mu}_2)^2, \\
\widehat{K}_{\text{local}} &:= \max\left(\widehat{\sigma}_U^2,\, \widehat{\sigma}_V^2\right) + 2\max\left(|\overline{U} - \widehat{\mu}_1|,\, |\overline{V} - \widehat{\mu}_2|\right) + 1 \geq 1.
\end{aligned}$$

Definition 5.3 (Curvature Constant M_t and Shrinking Ball B_t)**.** At round t, define the data-dependent curvature constant M_t and Euclidean ball B_t of radius $r_t := 1/\sqrt{M_t t}$ around $\widehat{\theta}$ by

$$M_t := \frac{16\widehat{K}_{\text{local}}}{\epsilon_0^3} + \frac{8\widehat{\Delta}_{\text{mean}} + 2}{\epsilon_0^2} + 1, \qquad B_t := \left\{ \theta' \in \Theta : \|\theta' - \widehat{\theta}\|_2 \leq r_t = \frac{1}{\sqrt{M_t t}} \right\}. \tag{29}$$

Multiplying M_t by ϵ_0^2 and using $\widehat{K}_{\text{local}} \geq 1$ and $\widehat{\Delta}_{\text{mean}} \geq 1$ gives $M_t \epsilon_0^2 \geq 8\widehat{\Delta}_{\text{mean}} + 2 \geq 10$, which implies $M_t \geq \frac{10}{\epsilon_0^2} \geq 40$ for all $t \geq 1$.

Lemma 5.4 (Gershgorin Bound on H_t)**.** *For every round $t \geq 1$ and every $\theta' = (\mu_1', \mu_2', \rho') \in B_t$:*

1. $r_t \leq \frac{\epsilon_0}{2}$ *and* $r_t \leq \frac{1}{\sqrt{40}} < \frac{1}{6} < 1$;
2. $B_t \subset \Theta = \mathbb{R}^2 \times (-1, 1)$, $1 - \rho'^2 \geq \epsilon_0$, *and* $|\mu_i' - \widehat{\mu}_i| \leq r_t < \frac{1}{6}$ *for* $i \in \{1, 2\}$;
3. $H_t(\theta') \preceq M_t t \, I_{3\times 3}$.

Proof. **Step 1:** Since $M_t t \epsilon_0^2 \geq 10t \geq 10 > 4$ and $M_t t \geq 40$, we immediately obtain $r_t = 1/\sqrt{M_t t} \leq \epsilon_0/2$ and $r_t \leq 1/\sqrt{40} < 1/6$.

Step 2: Since $1 - \widehat{\rho}^2 = (1 - |\widehat{\rho}|)(1 + |\widehat{\rho}|) \leq 2(1 - |\widehat{\rho}|)$, we have $1 - |\widehat{\rho}| \geq \frac{1-\widehat{\rho}^2}{2} = \epsilon_0 > r_t$. Thus every point in the ball satisfies $|\rho'| \leq |\widehat{\rho}| + r_t < 1$, so $B_t \subset \Theta$. Furthermore, $|(1 - \rho'^2) - (1 - \widehat{\rho}^2)| = |\rho' - \widehat{\rho}||\rho' + \widehat{\rho}| \leq 2r_t \leq \epsilon_0$, so $1 - \rho'^2 \geq 2\epsilon_0 - \epsilon_0 = \epsilon_0$.

Step 3: On B_t, $|\overline{U} - \mu_1'| \leq |\overline{U} - \widehat{\mu}_1| + |\widehat{\mu}_1 - \mu_1'| \leq |\overline{U} - \widehat{\mu}_1| + 1 \leq \widehat{\Delta}_{\text{mean}}$,

$$\frac{1}{N_{1,2}} \sum_{j=1}^{N_{1,2}} (U_j - \mu_1')^2 = \widehat{\sigma}_U^2 + 2(\widehat{\mu}_1 - \mu_1')(\overline{U} - \widehat{\mu}_1) + (\widehat{\mu}_1 - \mu_1')^2 \leq \widehat{\sigma}_U^2 + 2|\overline{U} - \widehat{\mu}_1| + 1 \leq \widehat{K}_{\text{local}},$$

and by $|ab| \leq \frac{1}{2}(a^2 + b^2)$, $\frac{1}{N_{1,2}} \left| \sum_{j=1}^{N_{1,2}} (U_j - \mu_1')(V_j - \mu_2') \right| \leq \widehat{K}_{\text{local}}$. Substituting these into (25)–(28) with $N_1, N_2, N_{1,2} \leq t$, $|\rho'| < 1$, and $1 - \rho'^2 \geq \epsilon_0$ gives:

$$\frac{|H_{11}|}{t}, \frac{|H_{22}|}{t} \leq 1 + \frac{1}{\epsilon_0}, \quad \frac{|H_{12}|}{t} \leq \frac{1}{\epsilon_0}, \quad \frac{|H_{13}|}{t}, \frac{|H_{23}|}{t} \leq \frac{4\widehat{\Delta}_{\text{mean}}}{\epsilon_0^2}, \quad \frac{|H_{33}|}{t} \leq \frac{2}{\epsilon_0^2} + \frac{16\widehat{K}_{\text{local}}}{\epsilon_0^3}.$$

Since $\epsilon_0 \leq \frac{1}{2} \implies \frac{2}{\epsilon_0} \leq \frac{1}{\epsilon_0^2} \leq \frac{2}{\epsilon_0^2}$, all three absolute row sums of $\frac{1}{t} H_t(\theta')$ are bounded by M_t. By Gershgorin's Circle Theorem, $\lambda_{\max}(H_t(\theta')) \leq \max_i \sum_{j=1}^3 |H_{ij}| \leq M_t t$. □

5.3 Explicit Pathwise Bound on Δ_t via a Cauchy–Uniform Prior

To avoid any bounded-box assumption on (μ_1, μ_2) and eliminate $p_{\min,t}$, we equip $\Theta = \mathbb{R}^2 \times (-1, 1)$ with independent standard Cauchy priors on μ_1', μ_2' and a uniform prior on $\rho' \in (-1, 1)$:

$$p(\mu_1', \mu_2', \rho') = \frac{1}{2} \cdot \frac{1}{\pi(1 + \mu_1'^2)} \cdot \frac{1}{\pi(1 + \mu_2'^2)} = \frac{1}{2\pi^2(1 + \mu_1'^2)(1 + \mu_2'^2)}. \tag{30}$$

Theorem 5.4: Explicit Pathwise Bound on Δ_t and δ-PAC Stopping Threshold

Theorem 5.5 (Explicit Pathwise Bound on Δ_t and δ-PAC Stopping Threshold)**.** *Under the Cauchy–Uniform prior (30), for every round $t \geq 1$ and every dataset:*

$$\Delta_t \leq \frac{3}{2} \log t + \mathcal{C}_t, \qquad \textit{where} \qquad \mathcal{C}_t := \frac{3}{2} \log M_t + \log(1 + \widehat{\mu}_1^2) + \log(1 + \widehat{\mu}_2^2) + 3. \tag{31}$$

Consequently, stopping at $\tau_\delta = \inf\{t \geq 1 : Z_t > \beta(t, \delta)\}$ with threshold

$$\beta(t, \delta) := \log \frac{1}{\delta} + \frac{3}{2} \log t + \frac{3}{2} \log M_t + \log(1 + \widehat{\mu}_1^2) + \log(1 + \widehat{\mu}_2^2) + 3 \tag{32}$$

is unconditionally δ-PAC: $\mathbb{P}_\theta\big(\tau_\delta < \infty, \widehat{a}_{\tau_\delta} \neq \arg\max_{i \in \{1,2\}} \mu_i\big) \leq \delta$ for all $\theta \in \Theta$ with $\mu_1 \neq \mu_2$.

Proof. For any $\theta' \in B_t$, a second-order Taylor expansion around $\widehat{\theta}$ (where $\nabla \ell_t(\widehat{\theta}) = 0$) with Lagrange remainder at $\tilde{\theta} \in B_t$ gives $\ell_t(\widehat{\theta}) - \ell_t(\theta') = \frac{1}{2}(\theta' - \widehat{\theta})^\top H_t(\tilde{\theta})(\theta' - \widehat{\theta}) \leq \frac{M_t t}{2} r_t^2 = \frac{1}{2}$, so $L_t(\theta') \geq e^{-1/2} \sup_\Theta L_t$. Moreover, since $|\mu_i' - \widehat{\mu}_i| \leq r_t < \frac{1}{6}$ on B_t, using $\frac{1}{3}|\widehat{\mu}_i| \leq \frac{1}{6}\widehat{\mu}_i^2 + \frac{1}{6}$ gives

$$1 + \mu_i'^2 \leq 1 + \left(|\widehat{\mu}_i| + \frac{1}{6}\right)^2 = 1 + \widehat{\mu}_i^2 + \frac{1}{3}|\widehat{\mu}_i| + \frac{1}{36} \leq \frac{43}{36}(1 + \widehat{\mu}_i^2) < \frac{6}{5}(1 + \widehat{\mu}_i^2).$$

Thus $p(\theta') \geq \left[2\pi^2(\frac{6}{5})^2(1+\widehat{\mu}_1^2)(1+\widehat{\mu}_2^2)\right]^{-1}$ for all $\theta' \in B_t$. Restricting the mixture integral to $B_t \subset \Theta$, multiplying by $\mathrm{Vol}(B_t) = \frac{4\pi}{3(M_t t)^{3/2}}$, and taking negative logarithms yields (31) since $\log\left(\frac{3\pi}{2}(\frac{6}{5})^2 e^{1/2}\right) \approx 2.415 < 3$. Combining (31) with Theorem 5.2 establishes δ-PAC soundness. □

Lemma 5.6 (Deterministic Bound on M_t and $\mathcal{C}_t$ When Estimates Are Accurate)**.** *Fix $\eta \in \left(0, \frac{1-|\rho|}{2}\right]$ and let $\underline{\epsilon} := \frac{3}{8}(1-|\rho|) > 0$. Whenever $|\widehat{\rho}(t) - \rho| \leq \eta$, $|\widehat{\mu}_i(t) - \mu_i| \leq \eta$ $(i = 1, 2)$, $|\overline{U}(t) - \mu_1| \leq \eta$, $|\overline{V}(t) - \mu_2| \leq \eta$, and the sample variances of U and V are at most $1 + \eta$, we have*

$$M_t \leq \overline{M}(\rho, \eta) := \frac{16(2 + 5\eta + 4\eta^2)}{\underline{\epsilon}^3} + \frac{10 + 16\eta}{\underline{\epsilon}^2} + 1, \tag{33}$$

$$\mathcal{C}_t \leq C_\eta := \frac{3}{2}\log \overline{M}(\rho, \eta) + \log\big(1 + (|\mu_1| + \eta)^2\big) + \log\big(1 + (|\mu_2| + \eta)^2\big) + 3 < \infty. \tag{34}$$

Proof. Since $|\widehat{\rho}(t)| \leq |\rho| + \eta \leq \frac{1+|\rho|}{2}$, we have $1 - \widehat{\rho}(t)^2 \geq 1 - \left(\frac{1+|\rho|}{2}\right)^2 = \frac{(1-|\rho|)(3+|\rho|)}{4} \geq \frac{3}{4}(1-|\rho|)$, so $\epsilon_0 = \frac{1-\widehat{\rho}(t)^2}{2} \geq \frac{3}{8}(1-|\rho|) = \underline{\epsilon}$. By the triangle inequality, $|\overline{U}(t) - \widehat{\mu}_1(t)| \leq 2\eta$ and $|\overline{V}(t) - \widehat{\mu}_2(t)| \leq 2\eta$, so $\widehat{\Delta}_{\text{mean}} \leq 1 + 2\eta$. Furthermore, $\widehat{\sigma}_U^2 = \frac{1}{N_{1,2}}\sum(U_j - \overline{U})^2 + (\overline{U} - \widehat{\mu}_1)^2 \leq 1 + \eta + 4\eta^2$, so $\widehat{K}_{\text{local}} \leq 2 + 5\eta + 4\eta^2$. Substituting these into (29) and (31) gives (33) and (34). □

6 First-Order Asymptotic Cost Optimality

We now prove that our Cost-Weighted Correlated Track-and-Stop rule with joint-only forced exploration (Algorithm 1) is first-order asymptotically cost-optimal, following the proof strategy of Garivier and Kaufmann [2]. Because a single joint pull (U_j, V_j) observes all three parameters (μ_1, μ_2, ρ) simultaneously, forced exploration is applied only to the joint action $a = 12$.

Algorithm 1 Cost-Weighted Correlated Track-and-Stop

Require: confidence $\delta \in (0,1)$; individual pull cost $c > 0$ and joint pull cost $c_{1,2} > 0$

$N_1(0) \leftarrow 0$, $N_2(0) \leftarrow 0$, $N_{1,2}(0) \leftarrow 0$
for $t = 0, 1, 2, \ldots$ **do**
  Compute the MLE $\widehat{\theta}(t) = (\widehat{\mu}_1(t), \widehat{\mu}_2(t), \widehat{\rho}(t))$ and empirical leader $\widehat{a}_t = \arg\max_{i \in \{1,2\}} \widehat{\mu}_i(t)$
  Compute Z_t via (23) and $\beta(t, \delta)$ via (32)
  if $Z_t > \beta(t, \delta)$ **then**
    $\tau_\delta \leftarrow t$; **return** $\widehat{a}_t$
  end if
  if $N_{1,2}(t) < \sqrt{t+1}$ **then** ▷ Forced joint exploration
    Make a joint pull $(a_t = 12)$
  else if $I(\widehat{\theta}(t); v_J) > I(\widehat{\theta}(t); v_I)$ **then** ▷ Equivalent to $c_{1,2} < c_{1,2}^{\text{th}}(|\widehat{\mu}_1(t) - \widehat{\mu}_2(t)|, \widehat{\rho}(t))$
    Make a joint pull $(a_t = 12)$
  else if $I(\widehat{\theta}(t); v_I) > I(\widehat{\theta}(t); v_J)$ **then**
    if $N_1(t) \leq N_2(t)$ **then**
      pull Arm 1 $(a_t = 1)$
    else
      pull Arm 2 $(a_t = 2)$
    end if
  else
    Tie: choose either the joint pull or the alternating individual pull above
  end if
  Observe the outcome and increment $N_{a_t}(t+1) = N_{a_t}(t) + 1$
end for

6.1 Properties of $I(\theta; w)$

Allocation Simplex. The cost-normalized allocation simplex and the empirical allocation at round t are:

$$\mathcal{W} := \big\{w = (w_1, w_2, w_{12}) \in \mathbb{R}^3_{\geq 0} : cw_1 + cw_2 + c_{1,2}w_{12} = 1\big\}, \qquad w_a(t) := \frac{N_a(t)}{\mathrm{Cost}(t)}, \quad a \in \{1, 2, 12\}. \tag{35}$$

For any $w \in \mathcal{W}$, define:

$$I(\theta; w) := \inf_{\lambda \in \mathrm{Alt}_1} \big[w_1 D_1(\theta, \lambda) + w_2 D_2(\theta, \lambda) + w_{12} D_{12}(\theta, \lambda)\big], \qquad \mathcal{W}^\star := \arg\max_{w \in \mathcal{W}} I(\theta; w). \tag{36}$$

The distance of an allocation from $\mathcal{W}^\star$ is:

$$\mathrm{dist}(w, \mathcal{W}^\star) := \inf_{w^\star \in \mathcal{W}^\star} \|w - w^\star\|_1. \tag{37}$$

Lemma 6.1 (Concavity and Continuity of $I(\theta; w)$)**.** *The function $w \mapsto I(\theta; w)$ is concave and continuous on $\mathcal{W}$, $\mathcal{W}^\star$ is convex, and*

$$I^\star(\theta) \geq I(\theta; v_I) = \frac{(\mu_1 - \mu_2)^2}{8c} > 0. \tag{38}$$

The maps $\vartheta \mapsto I(\vartheta; v_I)$ and $\vartheta \mapsto I(\vartheta; v_J)$ are continuous on Θ. Moreover:

$$C^\star_\epsilon := \inf\Big\{I(\theta; w) : w \in \mathcal{W},\ \mathrm{dist}(w, \mathcal{W}^\star) \leq \epsilon\Big\} \longrightarrow I^\star(\theta) \qquad \text{as } \epsilon \downarrow 0. \tag{39}$$

Proof. **Concavity and continuity in w.** For each fixed λ, the map $w \mapsto w_1 D_1 + w_2 D_2 + w_{12} D_{12}$ is linear. $I(\theta; w)$ is the infimum of these linear functions, so it is concave on the polytope $\mathcal{W}$, and hence continuous on $\mathcal{W}$ [7, Theorem 10.2]. The set of maximizers of a concave function over a convex set is convex, so $\mathcal{W}^\star$ is convex.

Continuity in ϑ. $I(\vartheta; v_I) = \frac{(\vartheta_1 - \vartheta_2)^2}{8c}$ is continuous. For v_J, by (15):

$$I(\vartheta; v_J) = \frac{1}{2c_{1,2}} \inf_{\rho' \in (-1,1)} f(\rho'; \vartheta), \tag{40}$$

where f is jointly continuous in (ϑ, ρ') and tends to $+\infty$ as $\rho' \to \pm 1$. The infimum can therefore be taken over a compact interval of ρ', and Berge's Maximum Theorem gives continuity in ϑ.

Limit of $C^\star_\epsilon$. $I(\theta; \cdot)$ is continuous on the compact set $\mathcal{W}$, hence uniformly continuous. So for every $\gamma > 0$ there is $\epsilon > 0$ such that $\mathrm{dist}(w, \mathcal{W}^\star) \leq \epsilon$ implies $I(\theta; w) \geq I^\star(\theta) - \gamma$. Since also $C^\star_\epsilon \leq I^\star(\theta)$, we get $C^\star_\epsilon \to I^\star(\theta)$. □

6.2 Decomposition of $\ell_t(\theta) - \ell_t(\lambda)$ and the Good Event

Sample Counts. Assume $\mu_1 > \mu_2$ and $|\rho| < 1$. Line 8 of Algorithm 1 makes a joint pull whenever $N_{1,2}(t) < \sqrt{t+1}$, and Lines 13–14 alternate between Arm 1 and Arm 2 whenever an individual pull is chosen. Hence, at every round $t \geq 1$:

$$N_{1,2}(t) \geq \sqrt{t} - 1, \qquad |N_1(t) - N_2(t)| \leq 1, \tag{41}$$

and at most $\sqrt{t} + 1$ of the first t rounds are forced exploration rounds.

Sum and Difference Variables. For each joint pull $j \in \{1, \ldots, N_{1,2}(t)\}$, define:

$$W_j^+ := U_j + V_j \sim \mathcal{N}\big(\mu_1 + \mu_2,\, 2(1+\rho)\big), \qquad W_j^- := U_j - V_j \sim \mathcal{N}\big(\mu_1 - \mu_2,\, 2(1-\rho)\big). \tag{42}$$

Since $\mathrm{Cov}(U_j + V_j,\, U_j - V_j) = \mathrm{Var}(U_j) - \mathrm{Var}(V_j) = 0$ and the pair is jointly Gaussian, W_j^+ and W_j^- are independent. Their sample means are:

$$\overline{W}^+(t) = \frac{1}{N_{1,2}(t)} \sum_{j=1}^{N_{1,2}(t)} W_j^+, \qquad \overline{W}^-(t) = \frac{1}{N_{1,2}(t)} \sum_{j=1}^{N_{1,2}(t)} W_j^-. \tag{43}$$

Expansion of $\ell_t(\theta) - \ell_t(\lambda)$. Let θ be the true parameter and $\lambda = (\nu_1, \nu_2, \varrho) \in \Theta$ any candidate parameter. Using (22), $\ell_t(\theta) - \ell_t(\lambda)$ splits over the four independent terms X_1, X_2, W^+, W^- (with $N_i(t)\overline{X}_i(t) := 0$ if $N_i(t) = 0$):

$$\begin{aligned}
\ell_t(\theta) - \ell_t(\lambda) = N_1(t) &\left[\frac{(\mu_1 - \nu_1)^2}{2} + (\mu_1 - \nu_1)\big(\overline{X}_1(t) - \mu_1\big)\right] \\
&+ N_2(t) \left[\frac{(\mu_2 - \nu_2)^2}{2} + (\mu_2 - \nu_2)\big(\overline{X}_2(t) - \mu_2\big)\right] \\
&+ \frac{N_{1,2}(t)}{2}\Bigg[\log\frac{1+\varrho}{1+\rho} \\
&\quad + \left(\frac{1}{N_{1,2}(t)}\sum_{j=1}^{N_{1,2}(t)} \big(W_j^+ - (\mu_1+\mu_2)\big)^2\right)\left(\frac{1}{2(1+\varrho)} - \frac{1}{2(1+\rho)}\right) \\
&\quad + \frac{\big((\mu_1+\mu_2) - (\nu_1+\nu_2)\big)\big(\overline{W}^+(t) - (\mu_1+\mu_2)\big)}{1+\varrho} \\
&\quad + \frac{\big((\mu_1+\mu_2) - (\nu_1+\nu_2)\big)^2}{2(1+\varrho)}\Bigg] \\
&+ \frac{N_{1,2}(t)}{2}\Bigg[\log\frac{1-\varrho}{1-\rho} \\
&\quad + \left(\frac{1}{N_{1,2}(t)}\sum_{j=1}^{N_{1,2}(t)} \big(W_j^- - (\mu_1-\mu_2)\big)^2\right)\left(\frac{1}{2(1-\varrho)} - \frac{1}{2(1-\rho)}\right) \\
&\quad + \frac{\big((\mu_1-\mu_2) - (\nu_1-\nu_2)\big)\big(\overline{W}^-(t) - (\mu_1-\mu_2)\big)}{1-\varrho} \\
&\quad + \frac{\big((\mu_1-\mu_2) - (\nu_1-\nu_2)\big)^2}{2(1-\varrho)}\Bigg].
\end{aligned} \tag{44}$$

Shorthand. To keep the proofs readable, we write $a_i := \mu_i - \nu_i$ and $e_i := \overline{X}_i(t) - \mu_i$ for the individual arms, and for the joint pulls:

$$\begin{aligned}
d_\pm &:= a_1 \pm a_2, & e_\pm &:= \overline{W}^\pm(t) - (\mu_1 \pm \mu_2), \\
x_\pm &:= \frac{1 \pm \varrho}{1 \pm \rho}, & s_\pm &:= \frac{1}{2(1\pm\rho)N_{1,2}(t)}\sum_{j=1}^{N_{1,2}(t)} \big(W_j^\pm - (\mu_1 \pm \mu_2)\big)^2.
\end{aligned} \tag{45}$$

Since $\frac{1}{2(1\pm\varrho)} - \frac{1}{2(1\pm\rho)} = \frac{1}{2(1\pm\rho)}\big(\frac{1}{x_\pm} - 1\big)$, (44) becomes:

$$\begin{aligned}
\ell_t(\theta) - \ell_t(\lambda) = \sum_{i=1}^{2} N_i(t)\left[\frac{a_i^2}{2} + a_i e_i\right] \\
+ \frac{N_{1,2}(t)}{2}\sum_{\pm}\left[\log x_\pm + s_\pm\Big(\frac{1}{x_\pm} - 1\Big) + \frac{d_\pm e_\pm}{1\pm\varrho} + \frac{d_\pm^2}{2(1\pm\varrho)}\right].
\end{aligned} \tag{46}$$

The Good Event. Because only joint pulls are forced, $N_i(t)$ can stay small (for example when v_J is optimal). We therefore do not ask $\overline{X}_i(t)$ to be η-close to μ_i; we only ask $N_i(t)e_i^2 \le t^{1/4}$, which holds with high probability whatever the size of $N_i(t)$. Fix:

$$\eta_0 := \min\left\{\frac{1}{4},\ \frac{1-|\rho|}{2}\right\}. \tag{47}$$

For $\eta \in (0, \eta_0]$, the good event $\mathcal{R}_t(\eta)$ is the event on which:

1. **Individual pulls:** for each $i \in \{1, 2\}$ with $N_i(t) \geq 1$,

$$N_i(t)\big(\overline{X}_i(t) - \mu_i\big)^2 \leq t^{1/4}. \tag{48}$$

2. **Joint pull means:**

$$|\overline{U}(t) - \mu_1| \leq \eta, \qquad |\overline{V}(t) - \mu_2| \leq \eta, \qquad |e_+| \leq \eta, \qquad |e_-| \leq \eta. \tag{49}$$

3. **Joint pull second moments:**

$$s_+,\, s_- \in [1 - \eta,\, 1 + \eta]. \tag{50}$$

4. **Joint pull sample variances:**

$$\frac{1}{N_{1,2}(t)} \sum_{j=1}^{N_{1,2}(t)} \big(U_j - \overline{U}(t)\big)^2 \leq 1 + \eta, \qquad \frac{1}{N_{1,2}(t)} \sum_{j=1}^{N_{1,2}(t)} \big(V_j - \overline{V}(t)\big)^2 \leq 1 + \eta. \tag{51}$$

Since $\eta \leq \frac{1-|\rho|}{2}$, we also have:

$$\eta \leq 1 \pm \rho. \tag{52}$$

Lemma 6.2 (Concentration of $\mathcal{R}_t(\eta)$)**.** *For every $\eta \in (0, \eta_0]$, there exist constants $b_1, b_2 > 0$ depending only on η and ρ such that for all $t \geq 1$:*

$$\mathbb{P}_\theta\big(\mathcal{R}_t(\eta)^c\big) \leq b_1 t \exp\big(-b_2 t^{1/4}\big). \tag{53}$$

Proof. **Joint pulls.** By (41), $N_{1,2}(t) \in [\lceil\sqrt{t} - 1\rceil, t]$. Fix a deterministic count n in this range and consider the first n joint pulls. Each of $\overline{U}, \overline{V}, \overline{W}^{\pm}$ is Gaussian with variance at most $4/n$, so:

$$\mathbb{P}\big(|\overline{W}_n^{\pm} - (\mu_1 \pm \mu_2)| > \eta\big) \leq 2\exp\Big(-\frac{n\eta^2}{8}\Big) \leq 2\exp\Big(-\frac{\eta^2(\sqrt{t}-1)}{8}\Big), \tag{54}$$

and identically for $\overline{U}_n$ and $\overline{V}_n$. The scaled second moments

$$Q_n^{\pm} := \sum_{j=1}^{n} \frac{\big(W_j^{\pm} - (\mu_1 \pm \mu_2)\big)^2}{2(1 \pm \rho)} \sim \chi_n^2 \tag{55}$$

satisfy, by Lemma 1 of Laurent and Massart [5] with $x = n\eta^2/8$:

$$\mathbb{P}\big(|Q_n^{\pm} - n| > n\eta\big) \leq 2\exp\Big(-\frac{n\eta^2}{8}\Big). \tag{56}$$

The same bound applies to the sample variances of U and V.

Individual pulls. Fix $i \in \{1, 2\}$ and a deterministic count $n \in \{1, \ldots, t\}$. Since $\sqrt{n}\,(\overline{X}_{i,n} - \mu_i) \sim \mathcal{N}(0, 1)$:

$$\mathbb{P}\Big(n\big(\overline{X}_{i,n} - \mu_i\big)^2 > t^{1/4}\Big) = \mathbb{P}\big(|\mathcal{N}(0,1)| > t^{1/8}\big) \leq 2\exp\Big(-\frac{t^{1/4}}{2}\Big). \tag{57}$$

Union bound. Each random count takes at most t values. Summing the bounds above over all of them, for a constant $c_1 > 0$:

$$\mathbb{P}_\theta\big(\mathcal{R}_t(\eta)^c\big) \leq c_1 t \exp\Big(-\frac{\eta^2(\sqrt{t}-1)}{8}\Big) + 4t\exp\Big(-\frac{t^{1/4}}{2}\Big) \leq b_1 t \exp\big(-b_2 t^{1/4}\big), \tag{58}$$

where the last step uses $\sqrt{t} \geq t^{1/4}$. □

6.3 Consistency of the MLE, Growth of Z_t, and Tracking

Lemma 6.3 (Consistency of the MLE and Growth of Z_t)**.** *For every $\epsilon > 0$ and $\xi > 0$, there exist $\eta \in (0, \eta_0]$ and $t_0 \geq 1$ such that for all $t \geq t_0$, on $\mathcal{R}_t(\eta)$:*

1. *$\|\widehat{\theta}(t) - \theta\|_2 < \xi$, and $\widehat{\mu}_1(t) > \widehat{\mu}_2(t)$ whenever $\xi < \frac{\mu_1 - \mu_2}{2}$;*
2. *$Z_t \geq \mathrm{Cost}(t)\big(I(\theta; w(t)) - \epsilon\big)$, where $w_a(t) := N_a(t)/\mathrm{Cost}(t)$.*

Proof. We work on $\mathcal{R}_t(\eta)$ throughout and use the shorthand in (46).

Step 1 (Lower bound on each term). For any real a and e, completing the square gives:

$$\frac{a^2}{2} + ae = \frac{a^2}{4} + \Big(\frac{a}{2} + e\Big)^2 - e^2 \geq \frac{a^2}{4} - e^2. \tag{59}$$

Individual arms. Taking $a = a_i$, $e = e_i$, multiplying by $N_i(t)$ and using Condition 1:

$$N_i(t)\left[\frac{a_i^2}{2} + a_i e_i\right] \geq N_i(t)\frac{a_i^2}{4} - N_i(t)e_i^2 \geq N_i(t)\frac{a_i^2}{4} - t^{1/4}. \tag{60}$$

Joint pulls. Taking $a = d_\pm$, $e = e_\pm$ in (59) and dividing by $1 \pm \varrho$:

$$\frac{d_\pm e_\pm}{1 \pm \varrho} + \frac{d_\pm^2}{2(1 \pm \varrho)} \geq \frac{d_\pm^2}{4(1 \pm \varrho)} - \frac{e_\pm^2}{1 \pm \varrho}. \tag{61}$$

By Condition 2, $|e_\pm| \leq \eta$. Writing $1 \pm \varrho = x_\pm(1 \pm \rho)$ and using (52):

$$\frac{e_\pm^2}{1 \pm \varrho} \leq \frac{\eta^2}{(1 \pm \rho)\, x_\pm} \leq \frac{\eta}{x_\pm}. \tag{62}$$

By Condition 3, $s_\pm \in [1 - \eta, 1 + \eta]$, so:

$$s_\pm\Big(\frac{1}{x_\pm} - 1\Big) = \frac{s_\pm}{x_\pm} - s_\pm \geq \frac{1 - \eta}{x_\pm} - (1 + \eta). \tag{63}$$

Adding the last three displays, each $\pm$ bracket in (46) is at least:

$$\log x_\pm + \frac{1 - 2\eta}{x_\pm} - (1 + \eta) + \frac{d_\pm^2}{4(1 \pm \varrho)}. \tag{64}$$

Step 2 (Restriction to a compact set K). *(a) Means far from (μ_1, μ_2).* Since $1 \pm \varrho \leq 2$ and $d_+^2 + d_-^2 = (a_1 + a_2)^2 + (a_1 - a_2)^2 = 2a_1^2 + 2a_2^2$:

$$\frac{d_+^2}{4(1 + \varrho)} + \frac{d_-^2}{4(1 - \varrho)} \geq \frac{d_+^2 + d_-^2}{8} = \frac{a_1^2 + a_2^2}{4}. \tag{65}$$

The function $x \mapsto \log x + \frac{1-2\eta}{x}$ is minimized at $x = 1 - 2\eta$, so for $\eta \leq \frac{1}{4}$:

$$\log x_\pm + \frac{1 - 2\eta}{x_\pm} - (1 + \eta) \geq \log(1 - 2\eta) - \eta \geq \log\frac{1}{2} - \frac{1}{4} \geq -1. \tag{66}$$

Multiplying (64) by $\frac{N_{1,2}(t)}{2}$, summing over $\pm$ and adding (60):

$$\ell_t(\theta) - \ell_t(\lambda) \geq \Big(\frac{N_1(t)}{4} + \frac{N_{1,2}(t)}{8}\Big)a_1^2 + \Big(\frac{N_2(t)}{4} + \frac{N_{1,2}(t)}{8}\Big)a_2^2 - N_{1,2}(t) - 2t^{1/4}. \tag{67}$$

By (41), $N_i(t) \geq \frac{N_1(t) + N_2(t) - 1}{2} = \frac{t - N_{1,2}(t) - 1}{2}$, so for $t \geq 2$:

$$\frac{N_i(t)}{4} + \frac{N_{1,2}(t)}{8} \geq \frac{N_i(t) + N_{1,2}(t)}{8} \geq \frac{1}{8} \cdot \frac{t - 1 + N_{1,2}(t)}{2} \geq \frac{t - 1}{16} \geq \frac{t}{32}. \tag{68}$$

Hence, using $N_{1,2}(t) \le t$, for every $\lambda \in \Theta$:

$$\ell_t(\theta) - \ell_t(\lambda) \ge \frac{t}{32}\big(a_1^2 + a_2^2\big) - t - 2t^{1/4}. \tag{69}$$

Now take the point $\lambda^0 := \big(\frac{\mu_1+\mu_2}{2}, \frac{\mu_1+\mu_2}{2}, \rho\big) \in \mathrm{Alt}_1$. On $\mathcal{R}_t(\eta)$, (44) gives $\ell_t(\theta) - \ell_t(\lambda^0) \le C_0 t$ for a constant $C_0 > 0$ depending only on θ. Define:

$$M_\mu^2 := \max\Big\{32(C_0+4),\ \frac{(\mu_1-\mu_2)^2}{2}\Big\}. \tag{70}$$

If $a_1^2 + a_2^2 > M_\mu^2$, then (69) gives:

$$\ell_t(\theta) - \ell_t(\lambda) > (C_0+4)t - t - 2t^{1/4} \ge (C_0+1)t > \max\big\{0,\ \ell_t(\theta) - \ell_t(\lambda^0)\big\}. \tag{71}$$

Therefore $\ell_t(\lambda) < \ell_t(\theta)$ and $\ell_t(\lambda) < \ell_t(\lambda^0)$: such a λ is not the MLE, and since $\lambda^0 \in \mathrm{Alt}_1$, it is not needed for $\sup_{\mathrm{Alt}_1} \ell_t$.

(b) Correlations near ± 1. Let $a_1^2 + a_2^2 \le M_\mu^2$ and compare $\lambda = (\nu_1, \nu_2, \varrho)$ with $\tilde\lambda := (\nu_1, \nu_2, \rho)$. Both have the same means, so $\tilde\lambda \in \mathrm{Alt}_1$ if and only if $\lambda \in \mathrm{Alt}_1$, and the individual arm terms cancel:

$$\ell_t(\tilde\lambda) - \ell_t(\lambda) = \big[\ell_t(\theta) - \ell_t(\lambda)\big] - \big[\ell_t(\theta) - \ell_t(\tilde\lambda)\big] = \frac{N_{1,2}(t)}{2} \sum_{\pm} \Big(\big[\ \cdot\ \big]_\varrho - \big[\ \cdot\ \big]_\rho \Big), \tag{72}$$

where $[\,\cdot\,]_\varrho$ denotes the $\pm$ bracket of (46) at correlation ϱ. At $\varrho = \rho$ we have $x_\pm = 1$, so $[\,\cdot\,]_\rho$ stays bounded on $\mathcal{R}_t(\eta)$. By (64), $[\,\cdot\,]_\varrho \to +\infty$ as $\varrho \to \pm 1$, because $\log x + \frac{1-2\eta}{x} \to +\infty$ as $x \to 0$. Hence there is a constant $\delta_\rho \in (0, 1-|\rho|)$ depending only on θ such that:

$$|\varrho| > 1 - \delta_\rho \implies \ell_t(\lambda) < \ell_t(\tilde\lambda). \tag{73}$$

Conclusion of Step 2. Define the compact set:

$$K := \big\{\lambda \in \Theta : a_1^2 + a_2^2 \le M_\mu^2,\ |\varrho| \le 1 - \delta_\rho\big\}. \tag{74}$$

By (a) and (b), the MLE lies in K and $\sup_{\mathrm{Alt}_1} \ell_t = \sup_{\mathrm{Alt}_1 \cap K} \ell_t$. The choice of M_μ makes $\lambda^0 \in K$.

Step 3 ($\ell_t(\theta) - \ell_t(\lambda)$ is close to $\sum_a N_a(t) D_a(\theta,\lambda)$ on K). *Individual arms.* Since $D_i(\theta,\lambda) = \frac{a_i^2}{2}$, the individual term minus $N_i(t)D_i(\theta,\lambda)$ is $N_i(t)a_i e_i$. By Condition 1:

$$N_i(t)|e_i| = \sqrt{N_i(t)}\sqrt{N_i(t)e_i^2} \le \sqrt{t \cdot t^{1/4}} = t^{5/8}, \qquad \text{so} \qquad |N_i(t)a_i e_i| \le M_\mu t^{5/8}. \tag{75}$$

Joint pulls. Since $(U,V) \mapsto (W^+, W^-)$ is invertible and W^+, W^- are independent, D_{12} is the sum of the KL divergences of W^+ and W^-:

$$D_{12}(\theta,\lambda) = \frac{1}{2}\sum_{\pm}\left[\log x_\pm + \frac{1}{x_\pm} - 1 + \frac{d_\pm^2}{2(1\pm\varrho)}\right]. \tag{76}$$

This is the joint part of (46) with $s_\pm = 1$ and $e_\pm = 0$. So the joint part minus $N_{1,2}(t)D_{12}(\theta,\lambda)$ is:

$$\frac{N_{1,2}(t)}{2}\sum_{\pm}\left[(s_\pm - 1)\Big(\frac{1}{x_\pm} - 1\Big) + \frac{d_\pm e_\pm}{1\pm\varrho}\right]. \tag{77}$$

On K, $1 \pm \varrho \ge \delta_\rho$, so $\big|\frac{1}{x_\pm} - 1\big| = \frac{|\rho-\varrho|}{1\pm\varrho} \le \frac{2}{\delta_\rho}$; also $|d_\pm| \le \sqrt{d_+^2 + d_-^2} \le \sqrt{2}M_\mu$, and $|s_\pm - 1|, |e_\pm| \le \eta$. Each bracket is therefore at most $\eta\,\frac{2+\sqrt{2}M_\mu}{\delta_\rho}$ in absolute value. Combining both parts:

$$\sup_{\lambda\in K}\Big|\big(\ell_t(\theta) - \ell_t(\lambda)\big) - \big(N_1(t)D_1(\theta,\lambda) + N_2(t)D_2(\theta,\lambda) + N_{1,2}(t)D_{12}(\theta,\lambda)\big)\Big|$$

$$\leq 2M_\mu t^{5/8} + N_{1,2}(t)\,\eta\,C', \qquad \text{where} \qquad C' := \frac{2+\sqrt{2}M_\mu}{\delta_\rho}. \tag{78}$$

The first term comes from the individual arms and the second from the joint pulls.

Claim 1 (Consistency). Define:

$$K_\xi := \{\lambda \in K : \|\lambda - \theta\|_2 \geq \xi\}, \qquad \gamma_\xi := \min_{\lambda \in K_\xi} D_{12}(\theta, \lambda) > 0. \tag{79}$$

γ_ξ is positive because $D_{12}(\theta,\lambda) = 0$ only when $\lambda = \theta$ (a joint pull alone reveals μ_1, μ_2 and ρ), and K_ξ is compact. For the individual arms, $\frac{a^2}{2} + ae$ is minimized over a at $a = -e$, so by Condition 1:

$$N_i(t)\left[\frac{a_i^2}{2} + a_i e_i\right] \geq -\frac{N_i(t)e_i^2}{2} \geq -\frac{t^{1/4}}{2}. \tag{80}$$

For the joint pulls, the joint part of (78) gives at least $N_{1,2}(t)\big(D_{12}(\theta,\lambda) - \eta C'\big)$. Hence, for every $\lambda \in K_\xi$:

$$\ell_t(\theta) - \ell_t(\lambda) \geq N_{1,2}(t)\big(\gamma_\xi - \eta C'\big) - t^{1/4}. \tag{81}$$

Choosing $\eta \leq \frac{\gamma_\xi}{2C'}$ and using (41):

$$\ell_t(\theta) - \ell_t(\lambda) \geq \frac{\gamma_\xi}{2}\big(\sqrt{t} - 1\big) - t^{1/4} > 0 \qquad \text{for all large } t. \tag{82}$$

So no $\lambda \in K_\xi$ is the MLE, and by Step 2 no $\lambda \notin K$ is the MLE. Hence $\widehat{\theta}(t) \in K \setminus K_\xi$, that is, $\|\widehat{\theta}(t) - \theta\|_2 < \xi$. If $\xi < \frac{\mu_1 - \mu_2}{2}$, then $|\widehat{\mu}_i(t) - \mu_i| < \xi$ gives $\widehat{\mu}_1(t) > \widehat{\mu}_2(t)$.

Claim 2 (Growth of Z_t). By Claim 1, $\widehat{a}_t = 1$. Using $\ell_t(\widehat{\theta}(t)) \geq \ell_t(\theta)$ and Step 2:

$$Z_t = \ell_t(\widehat{\theta}(t)) - \sup_{\lambda \in \mathrm{Alt}_1} \ell_t(\lambda) \geq \ell_t(\theta) - \sup_{\lambda \in \mathrm{Alt}_1 \cap K} \ell_t(\lambda) = \inf_{\lambda \in \mathrm{Alt}_1 \cap K} \big[\ell_t(\theta) - \ell_t(\lambda)\big]. \tag{83}$$

By (78) and $N_{1,2}(t) \leq t$:

$$Z_t \geq \inf_{\lambda \in \mathrm{Alt}_1 \cap K} \sum_a N_a(t) D_a(\theta,\lambda) - 2M_\mu t^{5/8} - t\,\eta\,C'. \tag{84}$$

Since $\sum_a N_a(t) D_a(\theta,\lambda) = \mathrm{Cost}(t) \sum_a w_a(t) D_a(\theta,\lambda)$ and the infimum over a subset of Alt_1 is at least the infimum over Alt_1:

$$\inf_{\lambda \in \mathrm{Alt}_1 \cap K} \sum_a N_a(t) D_a(\theta,\lambda) \geq \mathrm{Cost}(t)\, I(\theta; w(t)). \tag{85}$$

Dividing the error by $\mathrm{Cost}(t) \geq c_{\min} t$:

$$\frac{2M_\mu t^{5/8} + t\,\eta\,C'}{\mathrm{Cost}(t)} \leq \frac{2M_\mu t^{-3/8} + \eta C'}{c_{\min}}. \tag{86}$$

Choosing $\eta \leq \frac{\epsilon c_{\min}}{2C'}$ and t_0 large enough that $2M_\mu t_0^{-3/8} \leq \frac{\epsilon c_{\min}}{2}$ (and that Claim 1 holds for $t \geq t_0$), the right-hand side is at most ϵ. Hence:

$$Z_t \geq \mathrm{Cost}(t)\big(I(\theta; w(t)) - \epsilon\big). \tag{87}$$

□

Choice of ξ. Fix $\epsilon > 0$. By Lemma 6.1, $\vartheta \mapsto I(\vartheta; v_I)$ and $\vartheta \mapsto I(\vartheta; v_J)$ are continuous. So when θ has a unique optimal vertex ($I(\theta; v_I) \neq I(\theta; v_J)$), we can choose $\xi \in \big(0, \min\{\frac{\mu_1-\mu_2}{2}, \eta_0\}\big)$ such that:

$$\|\vartheta - \theta\|_2 < \xi \implies \arg\max_{v\in\{v_I,v_J\}} I(\vartheta; v) = \arg\max_{v\in\{v_I,v_J\}} I(\theta; v). \tag{88}$$

Lemma 6.4 (Tracking the Optimal Allocation)**.** *If $\|\widehat{\theta}(s) - \theta\|_2 < \xi$ for all $s \in [\lceil t^{1/4}\rceil, t]$, then for a constant $C > 0$ depending only on c and $c_{1,2}$:*

$$\operatorname{dist}\big(w(t), \mathcal{W}^\star\big) \leq Ct^{-1/4}. \tag{89}$$

Proof. Among the first t rounds, the rounds before $\lceil t^{1/4}\rceil$ and the forced joint pulls number at most:

$$\lceil t^{1/4}\rceil + \sqrt{t} + 1 \leq 4t^{3/4}. \tag{90}$$

Every other round $s \geq \lceil t^{1/4}\rceil$ compares $I(\widehat{\theta}(s); v_I)$ with $I(\widehat{\theta}(s); v_J)$, and by (88) picks the vertex that is optimal for θ. We also use $\operatorname{Cost}(t) \geq c_{\min} t$ and $c\,(w_1(t) + w_2(t)) + c_{1,2}w_{12}(t) = 1$.

Only $v_J = (0, 0, \frac{1}{c_{1,2}})$ is optimal. Individual pulls happen only before round $\lceil t^{1/4}\rceil$, so:

$$N_1(t) + N_2(t) \leq \lceil t^{1/4}\rceil \leq 2t^{1/4} \implies w_1(t) + w_2(t) \leq \frac{2}{c_{\min}} t^{-3/4}. \tag{91}$$

Since $\big|w_{12}(t) - \frac{1}{c_{1,2}}\big| = \frac{c}{c_{1,2}}\big(w_1(t) + w_2(t)\big)$:

$$\|w(t) - v_J\|_1 \leq \Big(1 + \frac{c}{c_{1,2}}\Big)\frac{2}{c_{\min}} t^{-3/4} \leq Ct^{-1/4}. \tag{92}$$

Only $v_I = (\frac{1}{2c}, \frac{1}{2c}, 0)$ is optimal. Joint pulls happen only in the rounds counted above, so:

$$N_{1,2}(t) \leq 4t^{3/4} \implies w_{12}(t) \leq \frac{4}{c_{\min}} t^{-1/4}. \tag{93}$$

By (41), $|w_1(t) - w_2(t)| \leq \frac{1}{\operatorname{Cost}(t)} \leq \frac{1}{c_{\min} t}$, and $w_1(t) + w_2(t) = \frac{1 - c_{1,2}w_{12}(t)}{c}$. Writing $w_i(t) = \frac{w_1(t)+w_2(t)}{2} \pm \frac{w_1(t)-w_2(t)}{2}$:

$$\Big|w_i(t) - \frac{1}{2c}\Big| \leq \frac{c_{1,2}}{2c} w_{12}(t) + \frac{1}{2c_{\min} t} \implies \|w(t) - v_I\|_1 \leq \Big(1 + \frac{c_{1,2}}{c}\Big) w_{12}(t) + \frac{1}{c_{\min} t} \leq Ct^{-1/4}. \tag{94}$$

Both v_I and v_J are optimal. By Theorem 3.1, $\mathcal{W}^\star = \{w \in \mathcal{W} : w_1 = w_2\}$. The point $w' := \big(\frac{w_1(t)+w_2(t)}{2}, \frac{w_1(t)+w_2(t)}{2}, w_{12}(t)\big)$ lies in $\mathcal{W}^\star$, so:

$$\operatorname{dist}\big(w(t), \mathcal{W}^\star\big) \leq \|w(t) - w'\|_1 = |w_1(t) - w_2(t)| \leq \frac{1}{c_{\min} t} \leq Ct^{-1/4}. \tag{95}$$

□

6.4 Main Optimality Theorem

Theorem 6.5: First-Order Asymptotic Cost Optimality

Theorem 6.5 (First-Order Asymptotic Cost Optimality)**.** *For every $\theta = (\mu_1, \mu_2, \rho) \in \Theta$ with $\mu_1 \neq \mu_2$, Algorithm 1 is δ-PAC for all $\delta \in (0, 1)$ and matches the lower bound of Theorem 2.1 asymptotically:*

$$\lim_{\delta\downarrow 0} \frac{\mathbb{E}_\theta[\operatorname{Cost}(\tau_\delta)]}{\log(1/\delta)} = \frac{1}{I^\star(\theta)}. \tag{96}$$

Proof. **δ-PAC.** Algorithm 1 stops with the threshold (32), so it is δ-PAC by Theorem 5.5, whatever the sampling rule.

Choice of constants. Fix $\epsilon > 0$ small enough that $\widetilde{C}_\epsilon := C^\star_\epsilon - \epsilon > 0$, fix ξ as in (88) (so $\xi \le \eta_0$), and let $\eta \in (0, \eta_0]$ and t_0 be given by Lemma 6.3 for this ϵ and ξ.

The good event fails only finitely often. Define:

$$\mathcal{G}_t := \bigcap_{s=\lceil t^{1/4} \rceil}^{t} \mathcal{R}_s(\eta), \qquad T_{\text{bad}} := \sup\{t \ge 1 : \mathcal{G}_t \text{ fails}\} \quad (\sup \emptyset = 0). \tag{97}$$

For each $s \in [\lceil t^{1/4} \rceil, t]$, we have $s \le t$ and $s^{1/4} \ge t^{1/16}$, so Lemma 6.2 gives $\mathbb{P}_\theta(\mathcal{R}_s(\eta)^c) \le b_1 s \exp(-b_2 s^{1/4}) \le b_1 t \exp(-b_2 t^{1/16})$. There are at most t such s, so:

$$\mathbb{P}_\theta(\mathcal{G}_t^c) \le b_1 t^2 \exp\big(-b_2 t^{1/16}\big). \tag{98}$$

Since $\{T_{\text{bad}} \ge n\} \subseteq \bigcup_{t \ge n} \mathcal{G}_t^c$:

$$\mathbb{E}_\theta[T_{\text{bad}}] = \sum_{n=1}^{\infty} \mathbb{P}_\theta(T_{\text{bad}} \ge n) \le \sum_{n=1}^{\infty} \sum_{t \ge n} \mathbb{P}_\theta(\mathcal{G}_t^c) = \sum_{t=1}^{\infty} t\, \mathbb{P}_\theta(\mathcal{G}_t^c) \le \sum_{t=1}^{\infty} b_1 t^3 \exp\big(-b_2 t^{1/16}\big) < \infty. \tag{99}$$

Choose a deterministic $T_1 \ge 1$ with $T_1^{1/4} \ge t_0$ and $C T_1^{-1/4} \le \epsilon$, and set:

$$T_\epsilon := \max\big(T_{\text{bad}} + 1,\, T_1\big), \qquad \mathbb{E}_\theta[T_\epsilon] \le \mathbb{E}_\theta[T_{\text{bad}}] + 1 + T_1 < \infty. \tag{100}$$

After T_ϵ. For every $t \ge T_\epsilon$, $\mathcal{G}_t$ holds and every $s \in [\lceil t^{1/4} \rceil, t]$ satisfies $s \ge t_0$. Hence:

1. By Lemma 6.3, for all $s \in [\lceil t^{1/4} \rceil, t]$:

$$\|\widehat{\theta}(s) - \theta\|_2 < \xi \le \eta_0, \qquad \widehat{\mu}_1(s) > \widehat{\mu}_2(s). \tag{101}$$

2. By Lemma 6.4 and the definition of $C^\star_\epsilon$ in Lemma 6.1:

$$\operatorname{dist}\big(w(t), \mathcal{W}^\star\big) \le C t^{-1/4} \le \epsilon \implies I(\theta; w(t)) \ge C^\star_\epsilon. \tag{102}$$

3. By Lemma 6.3:

$$Z_t \ge \operatorname{Cost}(t)\big(I(\theta; w(t)) - \epsilon\big) \ge \operatorname{Cost}(t)\big(C^\star_\epsilon - \epsilon\big) = \operatorname{Cost}(t)\, \widetilde{C}_\epsilon. \tag{103}$$

4. By Lemma 5.6 applied with η_0 (its conditions hold since $|\widehat{\rho}(t) - \rho|, |\widehat{\mu}_i(t) - \mu_i| < \xi \le \eta_0$, and Conditions 2 and 4 of $\mathcal{R}_t(\eta)$ hold with $\eta \le \eta_0$):

$$\mathcal{C}_t \le C_{\eta_0} < \infty. \tag{104}$$

Stopping. Since $t \le \operatorname{Cost}(t)/c_{\min}$, for $t \ge T_\epsilon$ the threshold satisfies:

$$\beta(t, \delta) = \log \frac{1}{\delta} + \frac{3}{2} \log t + \mathcal{C}_t \le \log \frac{1}{\delta} + \frac{3}{2} \log \frac{\operatorname{Cost}(t)}{c_{\min}} + C_{\eta_0}. \tag{105}$$

Define:

$$B_{\delta,\epsilon} := \inf\left\{ B \ge c_{\min} :\ x\, \widetilde{C}_\epsilon > \log \frac{1}{\delta} + \frac{3}{2} \log \frac{x}{c_{\min}} + C_{\eta_0} \ \text{ for all } x \ge B \right\}. \tag{106}$$

For any $t \ge T_\epsilon$ with $\operatorname{Cost}(t) \ge B_{\delta,\epsilon}$, items 3 and 4 give:

$$Z_t \ge \operatorname{Cost}(t)\, \widetilde{C}_\epsilon > \beta(t, \delta), \tag{107}$$

so the algorithm has stopped. Since the cost grows by at most $c_{\max}$ per round:

$$\mathrm{Cost}(\tau_\delta) \le \mathrm{Cost}(T_\epsilon) + B_{\delta,\epsilon} + c_{\max}. \tag{108}$$

Taking limits. Taking expectations and using $\mathbb{E}_\theta[\mathrm{Cost}(T_\epsilon)] \le c_{\max}\mathbb{E}_\theta[T_\epsilon] < \infty$ and $\limsup_{\delta\downarrow 0} \frac{B_{\delta,\epsilon}}{\log(1/\delta)} \le \frac{1}{\widetilde{C}_\epsilon}$:

$$\limsup_{\delta\downarrow 0} \frac{\mathbb{E}_\theta[\mathrm{Cost}(\tau_\delta)]}{\log(1/\delta)} \le \frac{1}{\widetilde{C}_\epsilon}. \tag{109}$$

Letting $\epsilon \downarrow 0$, $\widetilde{C}_\epsilon \to I^\star(\theta)$ by Lemma 6.1, so:

$$\limsup_{\delta\downarrow 0} \frac{\mathbb{E}_\theta[\mathrm{Cost}(\tau_\delta)]}{\log(1/\delta)} \le \frac{1}{I^\star(\theta)}. \tag{110}$$

Combining this with the lower bound $\liminf_{\delta\downarrow 0} \frac{\mathbb{E}_\theta[\mathrm{Cost}(\tau_\delta)]}{\log(1/\delta)} \ge \frac{1}{I^\star(\theta)}$ of Theorem 2.1 completes the proof. □

7 Numerical Experiments and Empirical Validation

In this section, we present three sets of numerical experiments: the phase transition of Theorem 3.1, joint pulls costing more than two individual pulls, and a comparison against existing baselines. Throughout all experiments, the individual pull cost is normalized to $c = 1$, and empirical expectations and 95% confidence intervals are computed over 1,000 independent Monte Carlo runs unless stated otherwise. Following the experimental protocol of Garivier and Kaufmann [2], every algorithm uses the exploration rate $\beta(t,\delta) = \log\big((\log t + 1)/\delta\big)$ in place of the theoretical threshold (32). In our implementation, matching Algorithm 1, forced exploration is applied to joint pulls only (a joint pull is forced whenever $N_{1,2}(t) < \sqrt{t+1}$, or $(t+1)^{1/3}$ for the second schedule), and individual pulls alternate between the two arms.

7.1 Experiment 1: Asymptotic Cost Ratio Across Correlation ρ and Joint Pull Cost $c_{1,2}$

Figure 4 compares the empirical expected stopping cost of our Cost-Weighted Correlated Track-and-Stop algorithm (which adaptively chooses between joint pulls and individual pulls) against the expected stopping cost of the standard no-joint-pull model across $\rho \in [-0.8, 0.8]$ for three joint pull costs: $c_{1,2} = 1.5$, $c_{1,2} = 2$, and $c_{1,2} = 3$, at both moderate confidence ($\delta = 0.1$) and high confidence ($\delta = 10^{-6}$), with mean gap $\mu_1 - \mu_2 = 0.2$[cite: 1]. This experiment uses 150 runs per configuration for our algorithm and 600 runs for the no-joint-pull baseline.

- Across all three panels ($c_{1,2} = 1.5, 2, 3$), both the moderate-confidence ($\delta = 0.1$) and asymptotic-regime ($\delta = 10^{-6}$) curves closely track the lower-bound ratio $I(\theta; v_I)/I^\star(\theta)$ (black dashed line)[cite: 1].
- Below the critical correlation threshold where $c_{1,2} > c^{\mathrm{th}}_{1,2}(\delta_0, \rho)$, the ratio stays flat at 1.0 (allowing joint pulls does not increase the cost asymptotically, because the algorithm then tracks v_I)[cite: 1]. Once ρ exceeds the phase-transition boundary, the cost ratio drops sharply toward 0 as $\rho \to 1$[cite: 1].

7.2 Experiment 2: Joint Pulls Costing More Than Two Individual Pulls $(2 < c_{1,2} < c^\star_{1,2})$

A counterintuitive consequence of Theorem 3.1 is that when the correlation $\rho > 0$ is sufficiently strong, joint pulls remain strictly optimal even when a single joint pull costs *more than pulling both arms individually* ($2c < c_{1,2} < c^\star_{1,2}$), because common-mode noise cancellation in $U_j - V_j \sim$

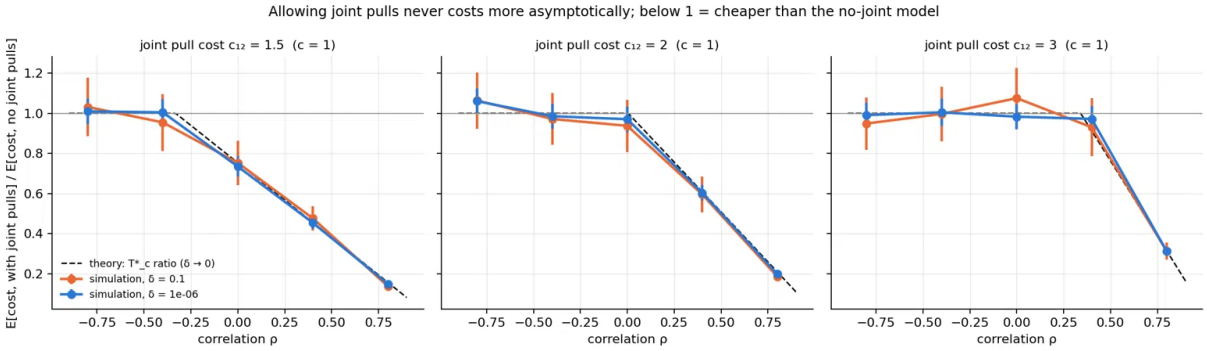


Figure 4: Ratio of expected stopping cost with joint pulls allowed to expected stopping cost without joint pulls ($\mathbb{E}[\text{Cost}_{\text{joint}}]/\mathbb{E}[\text{Cost}_{\text{no-joint}}]$) as a function of correlation $\rho \in [-0.8, 0.8]$ for $c_{1,2} \in \{1.5, 2, 3\}$ ($c = 1$, $\mu_1 - \mu_2 = 0.2$)[cite: 1]. The black dashed curve shows the theoretical asymptotic lower-bound ratio $I(\theta; v_I)/I^\star(\theta)$ as $\delta \to 0$, while the orange and blue curves show empirical simulation ratios at $\delta = 0.1$ and $\delta = 10^{-6}$[cite: 1]. Values below 1.0 indicate strict cost savings over the no-joint-pull model[cite: 1].

$\mathcal{N}(\mu_1 - \mu_2, 2(1-\rho))$ reduces the variance of the gap estimate by a factor of $1-\rho$. Figure 5 sweeps $c_{1,2}$ on two representative instances:

1. **Left column:** $\theta = (\mu_1, \mu_2, \rho) = (0.1, 0, 0.8)$, where the theoretical phase-transition threshold is $c^\star_{1,2} = 9.88$ (tested at $\delta = 0.1$ and $\delta = 0.01$).
2. **Right column:** $\theta = (\mu_1, \mu_2, \rho) = (0.2, 0, 0.5)$, where the theoretical phase-transition threshold is $c^\star_{1,2} = 3.93$ (tested at $\delta = 0.1$ and $\delta = 0.01$).

- For $\theta = (0.1, 0, 0.8)$ at $c_{1,2} = 2$, standard Garivier–Kaufmann Track-and-Stop (no joint pulls) requires an expected cost of 2,914 at $\delta = 0.1$ and 4,946 at $\delta = 0.01$, whereas our algorithm stops at an expected cost of 545 (saving 2,369, or 81.3%) at $\delta = 0.1$ and 981 (saving 3,966, or 80.2%) at $\delta = 0.01$.
- As $c_{1,2}$ increases across the shaded region, the stopping cost grows linearly with $c_{1,2}$ (matching the slope of the joint lower bound) until $c_{1,2}$ crosses $c^\star_{1,2} = 9.88$ (left) and $c^\star_{1,2} = 3.93$ (right), after which the algorithm switches to individual pulls and plateaus at the no-joint-pull baseline.

7.3 Experiment 3: Benchmark Against Existing Baselines Across Four Regimes

Figure 6 and Table 1 compare our algorithm (with both $\sqrt{t}$ and $t^{1/3}$ forced exploration) against five baselines—**GK Track-and-Stop** [no joint pulls; 2], **Top-Two Thompson Sampling** [8], **KL-LUCB** [3], **Joint pulls with ρ ignored**, and **Successive Elimination** [1]—across four canonical regimes (with $c = 1$; $\mu_1 - \mu_2 = 0.1$ in Regime A and $\mu_1 - \mu_2 = 0.2$ in Regimes B–D):

- **Regime A ($\rho = 0.8$, $c_{1,2} = 2$):** High positive correlation; joint pulls strongly reduce variance and lower the information-theoretic cost bound from 1,406 to 285 ($\delta = 0.1$) and from 3,603 to 729 ($\delta = 0.01$).
- **Regime B ($\rho = -0.5$, $c_{1,2} = 2$):** Negative correlation with $c_{1,2} = 2$; joint pulls do not help ($c_{1,2} > c^{\text{th}}_{1,2}$), and our algorithm automatically avoids joint pulls to match the optimal individual-pull baseline.
- **Regime C ($\rho = 0$, $c_{1,2} = 1.5$):** Uncorrelated arms with discounted joint pulls ($c_{1,2} = 1.5 < 2$); joint pulls reduce cost purely via the 25% measurement discount.

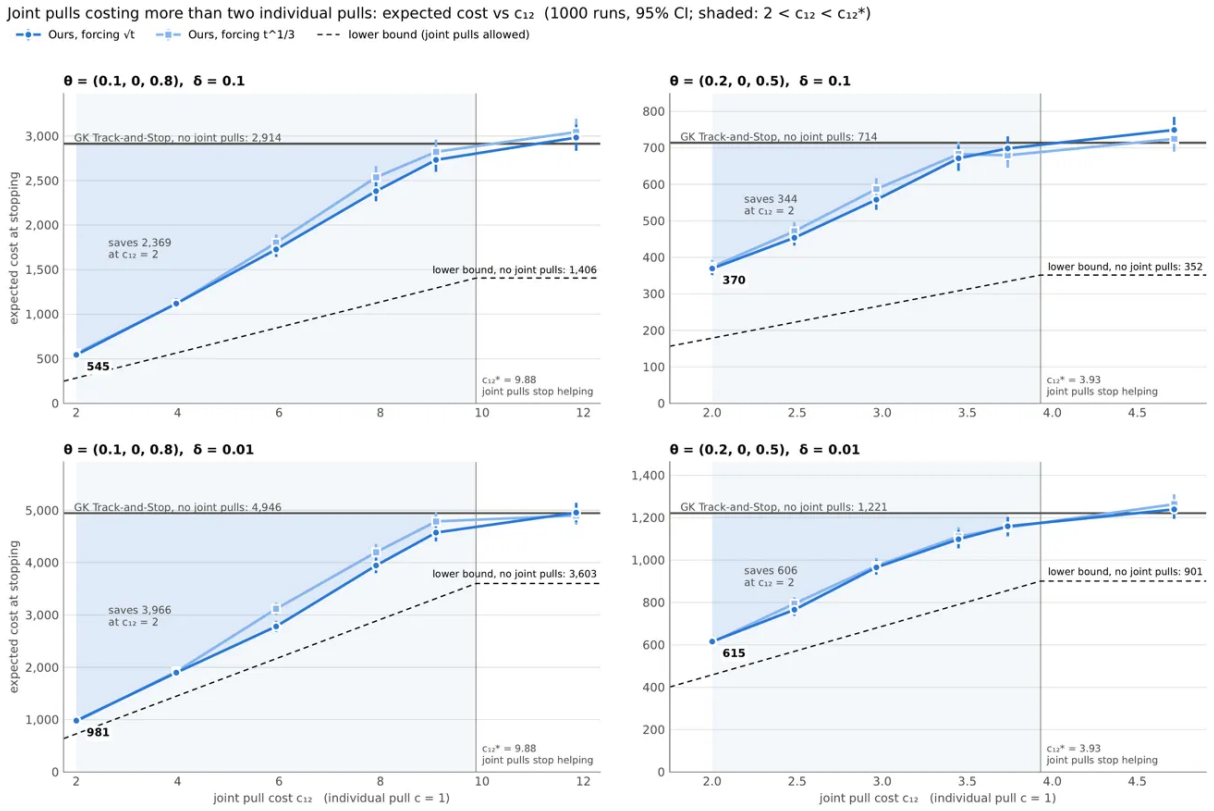


Figure 5: Expected cost at stopping versus joint pull cost $c_{1,2}$ ($c = 1$, 1,000 runs, 95% CI). The shaded region highlights the regime $2 < c_{1,2} < c^\star_{1,2}$ where a joint pull is more expensive than two separate individual pulls ($c_{1,2} > 2$), yet our algorithm has a much lower cost than standard Garivier–Kaufmann Track-and-Stop without joint pulls (horizontal solid line). The dashed curve is the lower bound with joint pulls allowed. Both forced-exploration schedules ($\sqrt{t}$ and $t^{1/3}$) are shown.

- **Regime D ($\rho = -0.5$, $c_{1,2} = 1.5$):** Negative correlation offsets the cost discount ($c_{1,2} = 1.5 > c_{1,2}^{\text{th}}$); our algorithm detects that negative correlation inflates the difference variance $2(1-\rho) = 3.0$ and sticks to individual pulls, whereas naive joint pulling fails.

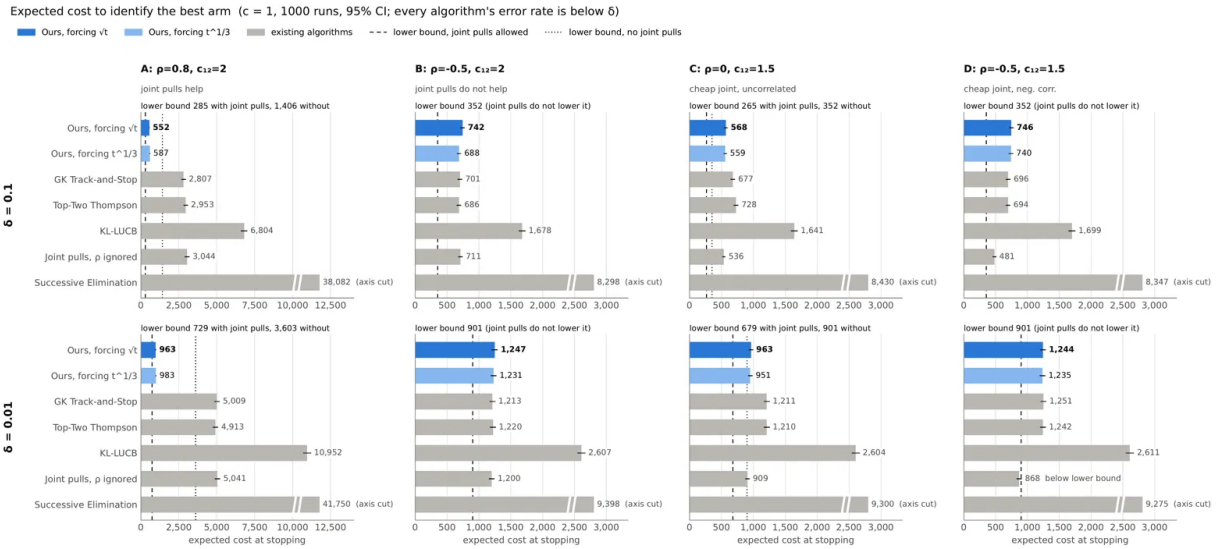


Figure 6: Expected cost to identify the best arm across Regimes A, B, C, and D at $\delta = 0.1$ (top row) and $\delta = 0.01$ (bottom row) over 1,000 runs (95% CI). Every algorithm's empirical error rate is below δ. Vertical dashed lines show the information-theoretic lower bound with joint pulls allowed; vertical dotted lines show the lower bound without joint pulls. Bars that run off the axis are cut and labeled with their value.

Table 1: Summary of Expected Stopping Costs Across Regimes A–D ($c = 1$, 1,000 runs).

Algorithm / Bound	**Confidence $\delta = 0.1$**				**Confidence $\delta = 0.01$**			
	Reg. A ($\rho = 0.8$)	**Reg. B** ($\rho = -0.5$)	**Reg. C** ($\rho = 0$)	**Reg. D** ($\rho = -0.5$)	**Reg. A** ($\rho = 0.8$)	**Reg. B** ($\rho = -0.5$)	**Reg. C** ($\rho = 0$)	**Reg. D** ($\rho = -0.5$)
Lower Bound (joint allowed)	*285*	*352*	*265*	*352*	*729*	*901*	*679*	*901*
Lower Bound (no joint pulls)	*1,406*	*352*	*352*	*352*	*3,603*	*901*	*901*	*901*
Ours (forcing $\sqrt{t}$)	**552**	**742**	**568**	**746**	**963**	**1,247**	**963**	**1,244**
Ours (forcing $t^{1/3}$)	**587**	**688**	**559**	**740**	**983**	**1,231**	**951**	**1,235**
GK Track-and-Stop	2,807	701	677	696	5,009	1,213	1,211	1,251
Top-Two Thompson	2,953	686	728	694	4,913	1,220	1,210	1,242
KL-LUCB	6,804	1,678	1,641	1,699	10,952	2,607	2,604	2,611
Joint pulls, ρ ignored	3,044	711	536	481	5,041	1,200	909	868
Successive Elimination	38,082	8,298	8,430	8,347	41,750	9,398	9,300	9,275

- When $\rho = 0.8$ and $c_{1,2} = 2$, existing algorithms that cannot exploit correlated joint pulls (GK Track-and-Stop, Top-Two Thompson) require $\sim$ 2,800–2,950 cost at $\delta = 0.1$ and $\sim$ 4,900–5,000 at $\delta = 0.01$. Our algorithm achieves 552 ($\delta = 0.1$) and 963 ($\delta = 0.01$), a $> 5\times$ reduction.
- The heuristic baseline "Joint pulls, ρ ignored" completely misses the correlation gain in Regime A (3,044 at $\delta = 0.1$ and 5,041 at $\delta = 0.01$) because its stopping statistic does not divide by $1 - \widehat{\rho}^2$, and in Regime D ($\rho = -0.5, c_{1,2} = 1.5$) it under-estimates the true variance $2(1-\rho) = 3$, causing it to stop prematurely at 868, below the lower bound of 901. This does not contradict Theorem 2.1, because this baseline is not δ-PAC: in additional runs on Regime D its empirical error rate exceeds δ for $\delta \leq 10^{-5}$.

8 Conclusion

We studied best-arm identification with individual and joint pulls of two correlated Gaussian arms under a sampling-cost budget. The max–min lower bound admits an exact vertex solution with an explicit phase transition in the joint pull cost, which shows that strongly correlated joint measurements can be worth more than twice the price of an individual pull. The profiled GLR statistic has a closed form, a data-dependent mixture-martingale threshold makes the stopping rule δ-PAC, and a cost-weighted correlated Track-and-Stop algorithm is first-order asymptotically cost-optimal. Natural next steps are the K-armed problem, for which Section 3 gives the semi-definite formulation, and sharper lower-order terms in the stopping threshold.

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